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Purpose-built WAAM 316L benchmark walls for thermo-mechanical calibration and strain-based hardness prediction

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Abstract

Wire Arc Additive Manufacturing (WAAM) of 316L generates complex thermal cycles, distortion and work hardening that are difficult to validate in 3D. Hence, in this paper, we design, build and fully characterise two benchmark walls (two-layer and three-layer) expressly for calibrating and validating open-source thermo-mechanical simulations in MOOSE with emphasis on hardness prediction. The geometry combines unfinished passes, controlled overlap and steady-state regions so that a single build exercises transient and quasi-stationary behaviour. A multi-zone thermocouple layout (top and bottom) enables a robust thermal calibration that reproduces peak temperatures and cooling histories. Full-field 3D scanning yields distortion maps condensed into a normalised warping index; MOOSE reproduces longitudinal/transverse distortion with absolute errors ≤0.10 mm. Microhardness maps of two cross-sections are compared to hardness fields reconstructed from the simulated accumulated equivalent plastic strain using a single Tabor-type law. Across lines V1/V2/H1, the full J_2-invariant surrogate achieves MAE ≈ 6-10 HV (comparable to test scatter), while a reduced "normal-only" strain measure shows MAE ≈ 25-35 HV and misses overlap-band peaks by ~20-40 HV, demonstrating the need for the full deviatoric invariant in WAAM 316L. Unlike prior WAAM validation studies that consider either single tracks or thermal/distortion only, this work delivers combined thermal, distortion and hardness datasets on multi-pass walls and an open, element-birth MOOSE workflow that enables full-field 3D hardness mapping from a calibrated strain surrogate. The benchmarks expose how layer count and pass pattern modulate hardness via reheating and constraint, and provide a reusable testbed for heat-source, boundary-condition and hardness-law comparisons.

Keywords: Wire Arc Additive Manufacturing (WAAM), thermo-mechanical simulations, hardness and distortion prediction, benchmark specimen design, MOOSE Framework.

Highlights:

- WAAM 316L benchmarks validate thermal, distortion, and hardness together.
- Warping index quantifies distortion; MOOSE matches within ≤0.10 mm.

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- Thermocouples and MOOSE calibration reproduce measured thermal histories.
- J_2-based hardness maps match 3D trends; simple model misses contrasts.
- Strain-based J_2 hardness surrogate enables full 3D hardness mapping in WAAM.

1. Introduction

In the rapidly advancing field of metal Additive Manufacturing (AM), accurate thermo-mechanical simulations are becoming critical tools for optimizing process parameters, predicting distortion, hardness, residual stresses, and improving part quality. Wire Arc Additive Manufacturing (WAAM), a Directed Energy Deposition (DED) method, has gained significant attention due to its high deposition rates, ability to fabricate large-scale components, and potential for cost-effective production. However, due to the inherent complexity of the WAAM process, including high heat input, complex thermal cycles, and layer-by-layer anisotropy, numerical models often rely on simplified assumptions or empirical calibrations. To validate and calibrate such simulations, reliable experimental benchmark specimens are essential. These specimens must be carefully designed and fabricated under controlled conditions to provide trustworthy data for comparison against thermal, phase-field, and mechanical simulation outputs. At the core of the WAAM process is the interaction between the nozzle, the wire feed system, and the electric arc, which collectively govern the deposition and quality of the built part. The nozzle typically comprises a gas shielding system, a contact tip, and a wire feeder, all functioning simultaneously to establish a stable arc and provide consistent melting of the feedstock. As the wire is continuously fed through the nozzle, an arc is struck between the wire tip and the substrate or the previous layer, thereby generating a localized molten pool. The shielding gas, often argon or an argon mixture, protects the melt pool from atmospheric contamination, ensuring metallurgical integrity. This process closely mirrors the principles of Gas Metal Arc Welding (GMAW), from which WAAM is fundamentally derived. As such, a deep understanding of welding science, including arc stability, heat input, metal transfer modes, and weld pool dynamics, is critical for achieving reliable and high-quality additive manufacturing results. For instance, parameters such as travel speed, arc voltage, wire feed rate, and torch angle, all well-studied in welding literature, directly impact bead geometry, heat distribution, and layer bonding in WAAM.

Despite the growing body of research in WAAM modelling, relatively few studies have focused on the systematic design and manufacturing of benchmark components specifically intended for hardness and distortion prediction. Existing efforts often centre on the simulation methodology itself, such as the calibration of parameters in melt pool simulation by Knapp et al. [1], or benchmark multi-layer simulations for small additively manufactured metal parts by Patil et al. [2], with minimal emphasis on how the physical specimens used for these validations are actually produced. Initiatives like the NIST Additive Manufacturing Benchmark Test Series (AM-Bench) have begun to bridge this gap for Powder Bed Fusion (PBF) systems by offering standardized datasets and measurement protocols [3]. However, there remains a critical lack of clear guidelines and case studies for WAAM-specific benchmark specimens that take into account process parameters, geometrical considerations, and measurement accessibility. In particular, in the area of hardness prediction models, there are few studies on WAAM fabricated components, and those focussed on the hardness of the materials are often limited to presenting experimental data, rather than numerical simulation solutions for a full sample analysis. As an example, the work of Jin et al. [4], Horgar et al. [5], and Caballero et al. [6], show and analyse the measured hardness of Ni-Cr-Mo-based superalloy, AA5183 wire, and 17-4 PH SS, respectively, but do not present modelling tools for the

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prediction of hardness for future material and process development. This paper aims to address this gap by presenting a methodology for the design and manufacturing of benchmark components tailored for the calibration and validation of thermo-mechanical simulations in WAAM for hardness and distortion prediction, contributing to more reproducible and trustworthy manufacturing and modelling practices in metal AM. In this work, hardness is not treated as an isolated output, but as a practical local property linked to the thermo-mechanical history of the build. For fully austenitic WAAM 316L, where solid-state phase transformation is not expected to dominate the response, local hardness variations mainly reflect differences in accumulated plastic strain, reheating, and constraint. This makes hardness a useful proxy for local work hardening and, therefore, for local variations in flow strength. Unlike tensile tests, which provide bulk average values, microhardness mapping resolves changes at fusion boundaries, overlap regions, and free edges. A validated hardness surrogate is therefore valuable because it allows the thermo-mechanical model to predict full-field property gradients that are difficult and expensive to obtain experimentally.

A growing body of work has established the core ingredients of thermo-mechanical modelling for WAAM: transient heat conduction with moving volumetric heat sources of ellipsoidal type and temperature-dependent properties [7]; progressive element activation to introduce material in step with deposition and to avoid unreal increases in convective and radiative area [8]; constitutive choices that range from small-strain J_2 plasticity with temperature-dependent hardening to faster inherent-strain surrogates for distortion; and microstructure surrogates in which hardness is inferred from accumulated plastic strain in the absence of phase change for fully austenitic 316L [9]. Recent reviews emphasise that predictive accuracy depends on joint calibration of heat input, losses and activation, and that robust validation should rely on multi-sensor thermal data and full-field distortion rather than pointwise comparisons [10]. At the same time, single-track studies are valuable for thermal tuning but do not translate directly to multi-track or multi-layer builds, because overlap and reheating alter constraint, plastic strain and therefore hardness, and models tuned on single tracks often miss these multi-pass effects [11], [12]. Motivated by this gap, the present benchmarks with two and three layers are designed to probe how an added layer redistributes hardness through inter-layer bands and free-edge softening, how pass order and asymmetry affect a normalised warping index, and whether a single Tabor-type calibration remains valid across different layer counts [13], [14], [15]. Also, while Tabor-style hardness–strain correlations are widely used for wrought 316 steels, coefficients and controlling invariants for AM or WAAM 316L are not well documented. We therefore compare a simplified normal-strain surrogate with a full J_2-invariant measure to identify the minimal invariant set that reproduces overlap-band hardening and edge softening.

As a wrap up, the present benchmark approach improves on previous WAAM validation studies in three ways. First, it uses purpose-built wall geometries that combine steady-state, overlap, and asymmetric deposition features within a small number of specimens, rather than relying on single tracks or generic walls. Second, it validates not only thermal histories and distortion, but also local hardness distributions, which are directly relevant to local property variation in WAAM 316L. Third, the workflow is implemented in an open-source multiphysics framework with progressive element activation, making the methodology reproducible and extendable to future benchmark studies.

2. Thermomechanical Modelling of the WAAM Process

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Thermo-mechanical simulations were performed in the open-source MOOSE framework [16], a parallel finite-element environment for tightly coupled multiphysics. We used MOOSE's kernels for transient heat conduction and inelastic constitutive updates together with user-defined materials/objects to couple thermal and mechanical fields at scale. The computational domain comprises the solid plate (see the green elements in Figure 1) plus an inactive surrounding region named *air* (see the yellow elements in Figure 1). The mesh consists of 68154 hexahedral elements and 79720 nodes.

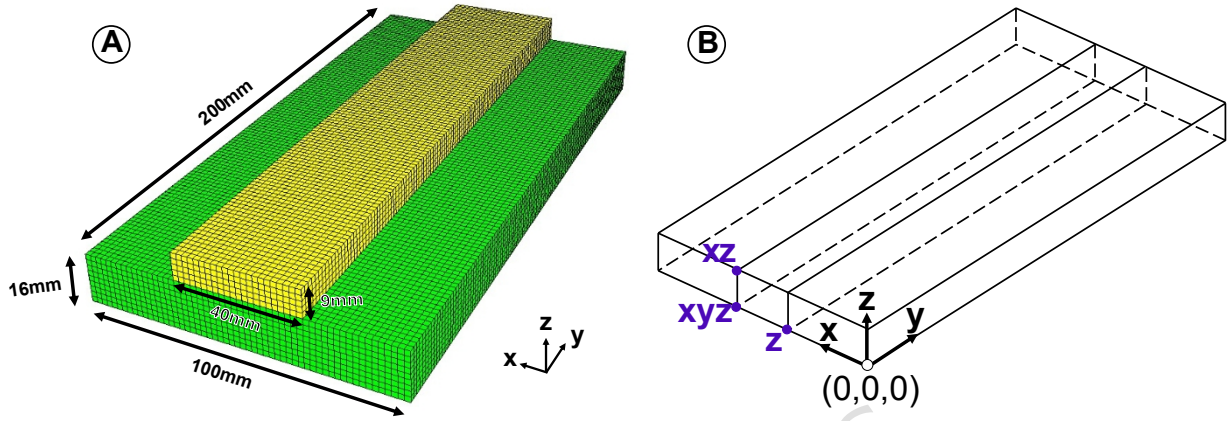


Figure 1: A) Model mesh depicting base plate (green elements) and air (yellow elements); B) Clamping conditions used for the simulations.

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2.1. Thermal Model

Thermal evolution follows the transient heat equation with a volumetric source (see Eq. 1):

$$\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (k \nabla T) + Q\ (x,y,z,t) \qquad \text{Eq. 1}$$

where:

- ρ is the density of the material (kg/m³)
- c_p is the specific heat capacity (J/kg·K)
- T is the temperature (K)
- t is time (s)
- k is the thermal conductivity (W/m·K)
- Q is the volumetric heat source (W/m³)

Convective and radiative losses are prescribed on exposed surfaces via a mixed heat-flux condition (see Eq. 2) at the boundary of the domain Ω:

$$-k \nabla T \cdot \hat{\boldsymbol{n}}|_{\partial\Omega} = h(T - T_\infty) + \sigma\varepsilon(T^4 - T_\infty^4) \qquad \text{Eq. 2}$$

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where:

- $\hat{\boldsymbol{n}}$ is the outward normal vector
- h is the convective heat transfer coefficient (W/m²·K)
- T_∞ is the ambient temperature (K)
- σ is the Stefan-Boltzmann constant ($5.67\ x\ 10^{-8}$ W/m²·K⁴)
- ε is the surface emissivity (dimensionless)

To improve robustness of the implicit time stepping, the nonlinear radiation term $\sigma\varepsilon(T^4 - T_\infty^4)$ is linearised about the ambient temperature T_∞. The transient analysis uses adaptive time stepping: small increments of the order of 0.05–0.1 s are taken during the rapid heating and cooling associated with the moving arc, while larger increments of up to 1–5 s are permitted during the slower inter-pass dwell.

2.2. Moving Volumetric Heat Source

Deposition heat input is represented by a translating 3D Gaussian/ellipsoidal volumetric source (see Eq. 3):

$$Q(x,y,z,t) = \frac{6\sqrt{3}Pf\eta}{abc\pi\sqrt{\pi}} \exp\left(-3\left[\frac{(x-x_t)^2}{a^2} + \frac{(y-y_t)^2}{b^2} + \frac{(z-z_t)^2}{c^2}\right]\right) \quad \text{Eq. 3}$$

where:

- P is the total power input from the arc (W)
- f is a scaling factor controlling the intensity of the heat source and to account for the mismatch between the idealised ellipsoidal source shape and the real arc energy distribution
- η is the process efficiency
- a,b,c are the semi-axis lengths of the ellipsoids in the x,y, and z directions (m)
- x,y,z are spatial coordinates with x aligned along the deposition path
- x_t,y_t,z_t are the instantaneous coordinates of the heat source centre at time t, which evolve according to the velocity vector and the initial scan positions (m)

2.3. Element-activation Strategy

Realistic WAAM simulation demands adding material only where and when it is deposited. Instead of activating a whole pass at room temperature, which unrealistically increases radiating area and biases peak temperatures, cooling rates, distortion, and residual stresses, we introduce material with a moving activation front that travels with the heat source (see Eq. 4). A binary indicator field $\chi(\boldsymbol{x},t) \in \{0,1\}$ marks whether solid material exists at a quadrature point (see Eq. 5). The substrate is initialized active ($\chi = 1$) and the yet-to-be-deposited region inactive ($\chi = 0$). A time-dependent ellipsoidal envelope $g(\boldsymbol{x},t)$ is centred at (x_t,y_t,z_t) with semi-axes (a,b,c), aligned to the scan direction, and evaluated through the normalized 3D Gaussian:

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$$g(\boldsymbol{x},t) = \frac{6\sqrt{3}}{abc\pi\sqrt{\pi}} exp\left(-3\left[\frac{(x-x_t)^2}{a^2} + \frac{(y-y_t)^2}{b^2} + \frac{(z-z_t)^2}{c^2}\right]\right) \quad \text{Eq. 4}$$

At each time step, a point is activated if $g(\boldsymbol{x},t) > \gamma$, with γ selecting a thin shell that matches the melt-pool footprint calibrated, and only active elements participate in the thermal and mechanical solutions; once χ becomes 1 it remains 1 for the rest of the analysis:

$$\chi(\boldsymbol{x},t_{n+1}) = \begin{cases} 1, & \text{if } g(\boldsymbol{x},t_{n+1}) > \gamma \\ \chi^n(\boldsymbol{x},t_n), & \text{if } g(\boldsymbol{x},t_{n+1}) \leq \gamma \end{cases} \quad \text{Eq. 5}$$

The same ellipsoid (a,b,c) and centre (x_t,y_t,z_t) parameterize the volumetric heat input, so the deposition footprint and energy input are consistent by construction. The semi-axes (a,b,c) denote the same physical quantities in Eq. (3) and Eq. (4): they are the ellipsoidal semi-axis lengths in the x, y and z directions, with x aligned along the deposition path. The heat-source term Eq. (3) and the activation envelope Eq. (4) share the same ellipsoid (a,b,c) and the same moving centre (x_t,y_t,z_t) by construction; the only difference is that Eq. (3) carries the energy scaling $(Pf\eta)$, whereas Eq. (4) is the normalised geometric envelope used solely to decide activation. The torch path is prescribed from start- and end-points and travel speed. The torch centre (x_t,y_t,z_t) advances continuously along the scan, and the next scan is triggered when the previously deposited region cools below a threshold temperature, reproducing inter-pass dwell. In practice this element-wise birth strategy stabilizes the thermal field by avoiding unphysical, instantaneous exposure of a whole bead to convection and radiation, keeps the evolving heat-loss area realistic, and improves residual-stress prediction by preserving the true sequence of thermal gradients and plastic straining. The approach remains general and fully geometry-agnostic: activation is controlled by a coupled field, not by hard-coded part geometry, so the same logic can be reused across tracks and layers. Its current scope assumes a sharp deposit front and perfect bonding at birth and uses a geometric envelope to decide activation; it does not explicitly model spatter, undercut, or partial fusion. Nonetheless, compared with whole-pass activation, the scan-wise strategy provides a closer match to WAAM physics and yields more accurate thermo-mechanical histories for stress and distortion analysis [17], [8].

Remelting of the previously deposited material and of the underlying substrate is not treated by element deactivation: once activated, an element remains active for the rest of the analysis. Instead, remelting is represented through the temperature dependence of the constitutive properties, the Young modulus E(T) collapses to a small residual value above the solidus (Table 1), so any region whose temperature exceeds the liquidus is physically the remelt region and contributes negligible load-bearing capacity to the mechanical system. The computed fusion zone (bounded by the T _liquidus isotherm) therefore comprises both freshly added wire material and re-melted pre-existing material, consistent with welding and WAAM physics. In MOOSE, element birth is implemented through the ActivateElementsCoupled object, which is the standard mechanism in the framework for progressively introducing material during a transient simulation. At each time step, ActivateElementsCoupled inspects the value of the coupled field χ(x, t) (driven by Eq. 5, which in turn is evaluated from g(x, t) in Eq. 4) and, for every element whose centroid

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lies within the active subdomain, switches the element from the inactive set to the active set. Newly activated elements are immediately included in the assembly of the residual and Jacobian for both the heat-conduction and the stress-equilibrium equations, so that the thermal and mechanical solutions automatically incorporate the new material from that step onward. Because activation is triggered by the coupled scalar field χ(x, t) rather than by a hard-coded element list, the same input deck applies to any deposition path and to any number of tracks or layers without modification. The thermal and mechanical problems are solved monolithically in a single coupled input file. The preconditioner is configured in "full" mode (SMP, full = true), so that MOOSE forms a single Jacobian matrix that is the standard derivative of the coupled residual of the heat-conduction equation and the stress-equilibrium equation with respect to all unknown fields (temperature, displacement components, and the plastic-strain internal variables). The mechanical contribution to the Jacobian is the consistent elasto-plastic tangent stiffness assembled from the active inelastic-strain models, which is the BlackBear default and is essential for quadratic Newton convergence in the presence of J2 plasticity. No staggered MultiApp decomposition is used; a single Newton solve advances temperature and displacement together at each adaptive time step.

2.4. Mechanical Model

The mechanical response of the WAAM 316L benchmark walls is described by a small strain thermo-elasto-plastic model with isotropic hardening, implemented in the open-source MOOSE framework. The mechanical problem is solved monolithically with the transient heat transfer analysis, so that temperature histories from the calibrated thermal model drive thermal strains and the temperature dependence of the elastic and plastic properties.

At each material point the total strain tensor (see Eq. 6) is additively decomposed into elastic, thermal and plastic parts:

$$\boldsymbol{\varepsilon}^{total} = \boldsymbol{\varepsilon}^{e} + \boldsymbol{\varepsilon}^{th} + \boldsymbol{\varepsilon}^{p} \qquad \text{Eq. 6}$$

and the Cauchy stress (see Eq. 7) follows isotropic linear elasticity with temperature dependent modulus and Poisson ratio:

$$\boldsymbol{\sigma} = \boldsymbol{C}(T) : (\boldsymbol{\varepsilon}^{total} - \boldsymbol{\varepsilon}^{th} - \boldsymbol{\varepsilon}^{p}) \qquad \text{Eq. 7}$$

Thermal strains are computed from the tabulated coefficient of thermal expansion of 316L by integrating from a reference temperature close to ambient. The body is only pinned at a few nodes to suppress rigid body motion, so that it is otherwise free to distort in response to the evolving thermal field (see Figure 1 B).

The mechanical boundary conditions require explanation, because the simulated and experimental restraints serve different purposes and represent two different instants of the process. During fabrication (Figure 2 and Figure 3 B) the base plate is held by four nickel-aluminium-bronze rods that clamp the two side edges. However, the distortion of interest, and the distortion that is measured by the GOM scan, is the residual, permanent distortion that develops after the part has cooled and the physical clamps have been released. Because even strong clamping cannot prevent the bending that springs back on unclamping, the released (post-clamp) state is the physically correct configuration to compare against the measurement. For this reason the baseline simulation uses a minimal statically determinate three-node pin set (Figure 1 B) that constrains exactly the six rigid-body modes of the plate (3 translations + 3

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rotations) without introducing any spurious elastic resistance to bending: the plate can therefore relax freely to whatever shape its locked-in plastic and thermal strains demand. This is precisely the state of the physical specimen the moment after the four bronze rods are released, when the only forces on the plate are gravity and its own internal residual stress. To verify this modelling choice quantitatively, an indirect check was carried out on the existing (pinned) simulation output, following the procedure proposed by an anonymous reviewer, rather than setting up a new coupled thermo-mechanical model. The transverse displacement history was extracted at two probe locations coincident with the physical clamp positions (span L ≈ 198 mm), and the differential displacement between them, Δu(t), was computed across the full deposition and cool-down history. The maximum differential displacement, Δu_max ≈ 0.039 mm, occurs at the end of the simulated history (T ≈ 290 K), corresponding to an imposed strain of ε ≈ 0.02%. Applying this displacement as an enforced elastic strain, using the model's own temperature-dependent elastic modulus E(T), gives an induced stress of σ_imposed ≈ 38 MPa. Comparing this against the model's own tabulated yield stress σy(T) at the corresponding temperature (σy ≈ 170 MPa near room temperature) shows that σ_imposed remains below 23% of yield throughout the entire deposition and cool-down history, with a minimum safety factor of 4.4. This estimate is itself conservative in two respects: it assumes the two ends are held perfectly rigid, rather than reflecting the finite compliance of the actual bronze clamp rods, and it uses a simple one-dimensional elastic estimate rather than resolving the full three-dimensional stress state; both simplifications tend to overstate σ_imposed relative to the true clamped condition. The constrained transverse expansion prevented by the physical end clamps therefore remains elastic throughout deposition and does not introduce additional plastic strain relative to the pinned baseline. Since elastic deformation fully recovers on unclamping, this indirect check confirms that the minimal-pin boundary condition correctly represents the released, post-clamp state measured by GOM.

Plasticity is modelled using standard J_2 (von Mises) theory with isotropic hardening. The deviatoric stress is $\boldsymbol{s} = \boldsymbol{\sigma} - \frac{1}{3}tr(\boldsymbol{\sigma})\mathbf{I}$, and the equivalent stress is:

$$\sigma_{eq} = \sqrt{\frac{3}{2}\boldsymbol{s} : \boldsymbol{s}} \qquad \text{Eq. 8}$$

Yielding (see Eq. 9) is governed by:

$$f(\boldsymbol{\sigma},\varepsilon_p,T) = \sigma_{eq} - \left[\sigma_y(T) + R(\varepsilon_p)\right] \leq 0 \qquad \text{Eq. 9}$$

where $\sigma_y(T)$ is the temperature dependent initial yield stress and $R(\varepsilon_p)$ is an isotropic hardening law specified as a tabulated function of the accumulated equivalent plastic strain ε_p. Associated flow is assumed, so the plastic strain rate is proportional to the gradient of the yield function, and the scalar measure ε_p is obtained by integrating the J_2 equivalent plastic strain rate over time. This same scalar is later used as the input to the strain-based hardness surrogate.

Temperature enters the constitutive response through two distinct routes, which it is important to distinguish. First, the temperature dependence of the yield surface is carried explicitly by $\sigma_y(T)$: the position of the yield surface (the value at zero plastic strain) is interpolated from the tabulated temperature-dependent yield stress at every quadrature point and time step, so that the material yields at progressively lower stress as temperature rises and effectively loses load-bearing capacity above the solidus. The elastic moduli $\boldsymbol{C}(T)$ and the thermal expansion are likewise temperature-dependent (Table

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1). Second, the strain-hardening increment $R(\varepsilon_p)$ is written as a function of accumulated plastic strain only, i.e. the shape of the hardening curve above yield is taken as temperature-independent. This is a deliberate modelling choice: the aim of the present study is the prediction of distortion and of post-build hardness, both of which are dominated by the thermal-mechanical history at moderate and intermediate temperatures (cooling, inter-pass dwell, reheating below the solidus) rather than by the detailed constitutive response of the very high-temperature material inside the molten pool. Because hardness and distortion are not strongly affected by how the hardening slope evolves at near-liquidus temperatures, the temperature dependence of $R(\varepsilon_p)$ is neglected and a single tabulated hardening curve is used; this is consistent with established practice in welding and WAAM thermo-mechanical simulation. The accumulated plastic strain ε_p produced under this formulation is therefore a physically meaningful measure of work hardening, which justifies its subsequent use as the input to the hardness surrogate; a fully temperature-dependent hardening table is identified as a straightforward refinement should higher-fidelity cyclic data become available.

The temperature dependent values used in the thermo-mechanical model are shown in Table 1, noting that the Poisson's ratio (v) is taken as 1 once the material reaches liquid state.

Table 1: Temperature-dependent material properties for WAAM.

T (K)	E (GPa)	v (–)	k (W/m·K)	c□ (J/kg·K)	α (×10⁻⁶/K)
293	195.6	0.29	14.12	492	14.6
373	191.2	0.29	15.26	502	15.4
473	185.7	0.29	16.69	514	16.2
573	179.6	0.29	18.11	526	16.9
673	172.6	0.29	19.54	538	17.4
773	164.5	0.29	20.96	550	17.8
873	155.0	0.29	22.38	562	18.1
973	144.1	0.29	23.81	575	18.4
1073	131.4	0.29	25.23	587	18.7
1173	116.8	0.29	26.66	599	19.0
1273	100.0	0.29	28.08	611	19.3
1373	80.0	0.29	29.50	623	19.5
1473	57.0	0.29	30.93	635	19.8
1573	30.0	0.29	32.35	647	20.0
1673	30.0	0.29	32.78	659	20.2

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The choice of an isotropic J_2 model is deliberate. It provides a simple and robust description of the dominant plasticity mechanisms in fully austenitic WAAM 316L, without introducing additional parameters related to back stress or texture. The same constitutive formulation has been previously calibrated and validated against residual stress measurements in WAAM 316L T-joints [18] and is reused here so that differences in the results arise from geometry and loading; not from changes in the mechanical model. In the present work, this isotropic formulation mainly serves to generate reliable fields of equivalent plastic strain, which are then post-processed into Vickers hardness maps and used to interpret the measured hardness distributions in the benchmark specimens.

The thermal and activation framework is not limited to 316L. In principle, the same workflow can be adapted to other wire-based DED or welding-based additive processes by updating the travel path, source calibration, and temperature-dependent material data. The present hardness law, however, is specific to fully austenitic 316L and would need to be recalibrated for alloys in which phase transformation contributes significantly to local hardness.

2.5. Hardness Surrogate Model

In the numerical simulations, Vickers hardness is not treated as an independent material property but is reconstructed a posteriori from the plastic strain field. The approach follows a Tabor style rationale in which hardness is assumed to scale with the accumulated equivalent plastic strain rather than with phase transformations, which is consistent with the fully austenitic solidification behaviour of 316L in this work. We proceed in two steps. First, we use a Simple Model (see Eq. 10) in which the equivalent plastic strain is computed from the normal components only:

$$\varepsilon_{eq}^{Simple} = \sqrt{\frac{2}{3}\left(\varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2\right)} \qquad \text{Eq. 10}$$

This provides a fast surrogate for hardness that captures the gross trends and is suitable for early calibration on 316L. We then adopt the Full Model (see Eq. 11). The underlying measure of plastic deformation in the Full Model is the accumulated equivalent plastic strain ε_{eq}^{Full}(t), obtained by time integrating the J_2 norm of the deviatoric plastic strain rate tensor. This scalar quantity excludes volumetric contributions and represents the full history of plastic straining at each material point, including both normal and shear components. In terms of the accumulated plastic strain components ε_{ij}, the final equivalent plastic strain at the end of the build can be written as:

$$\varepsilon_{eq}^{Full} = \sqrt{\frac{2}{3} max\left(0, \varepsilon_{xx}^2 + \varepsilon_{yy}^2 + \varepsilon_{zz}^2 + 2\left(\varepsilon_{xy}^2 + \varepsilon_{yz}^2 + \varepsilon_{xz}^2\right) - \frac{1}{3}\left(\varepsilon_{xx} + \varepsilon_{yy} + \varepsilon_{zz}\right)^2\right)} \qquad \text{Eq. 11}$$

where the $\max(0, \cdot)$-term is a numerical safeguard that avoids negative arguments of the square root due to round off. The local Vickers hardness HV (see Eq. 12) is then obtained for both models pointwise from a phenomenological power law of the form:

$$HV = A + B\left(\varepsilon_{eq}\right)^C \qquad \text{Eq. 12}$$

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The constants in Eq. 12 can be interpreted as an annealed base hardness (*A*), an amplitude (*B*) for strain induced hardening, and a strain sensitivity exponent (*C*), and were selected so that the predicted hardness range and gradients match the experimentally measured microhardness in the wall and heat affected zone. This functional form is compatible with classical Tabor type relations linking hardness to flow stress and accumulated plastic strain in metals, and with reported hardness-strain correlations for austenitic stainless steels [13], [14], [15]. Consistent with Tabor's proportionality between hardness and flow stress ($HV \approx 3\sigma$) [13], Eq. 12 may be viewed as $HV = HV_0 + \kappa\sigma_{flow}(\varepsilon_{eq})$ with $\kappa \approx 3$, in consistent units, so A represents the annealed base hardness and B lumps κ with the hardening strength, while C reflects the strain-hardening sensitivity (akin to a Hollomon exponent).

3. Experimental details

3.1. Materials

The material used for this study was 316L stainless steel, a corrosion-resistant alloy commonly utilized in civil engineering, marine applications, and biomedical industries. This steel is classified as an austenitic stainless steel, meaning it retains a fully austenitic microstructure across a wide range of temperatures. This stability is achieved through the presence of alloying elements such as nickel, manganese, and nitrogen, which prevent phase transformations during thermal cycles. Since no phase changes occur throughout the WAAM deposition process, this simplifies computational modelling and reduces the complexity of residual stress and distortion simulations. Furthermore, using a material with well-understood thermal and mechanical properties serves as a foundation for future research on alloys that undergo phase transformations due to temperature variations during additive manufacturing.

To ensure proper weld quality and minimize oxidation, argon (Ar) was used as the shielding gas during the WAAM process. Argon provides an inert atmosphere, preventing contamination and ensuring a stable arc. The filler material selected was 316LSi stainless steel wire, which contains a small percentage of silicon. The addition of silicon (Si) enhances arc stability and improves the fluidity of the molten weld pool, leading to better layer adhesion and surface finish. This characteristic is particularly beneficial when using Gas Metal Arc Welding (GMAW), also known as Metal Inert Gas (MIG) welding, the technique employed in this study. The Stainless Steel MIG Wire used in this study aligns with the AS/NZS ISO 14343-B ss316LSi and AWS/ASME-SFA A5.9 ER316LSi classifications [19]. The weight of the spool used was 15 kg and the wire diameter was 1.2 mm.

A complete table with 316L Stainless Steel composition for the base plate is shown below (Table 2).

Table 2: Composition of the 316L Stainless Steel used for the base plate.

Fe	Cr	Ni	Mo	Mn	Si	P	S	C
Bal.	16.0 – 18.0	10.0 – 14.0	2.0 – 3.0	2.0	1.0	0.045	0.015	0.03

Similarly, for the GMAW wire, the 316LSi Stainless Steel composition (as welded) is shown below (Table 3).

Table 3: Composition of the 316L Stainless Steel used for the GMAW wire (as welded).

Fe	Cr	Ni	Mo	Mn	Si	P	S	C	Cu
Bal.	18.08	11.40	2.57	1.90	0.71	0.12	0.008	0.017	0.18

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3.2. Design and Manufacturing Experimental Setup

To act as a benchmark and allow the calibration and validation of different parameters of the simulations, two samples with increasing number of layers were designed and manufactured. All the samples were manufactured at the University of Wollongong, at the Facility for Intelligent Fabrication (FIF), in collaboration with The University of Sydney and ANSTO. For the WAAM process in this study, a Fronius CMT 4000 Advanced power source system was used in combination with an ABB IRB 2600 robot [20]. This setup provided repeatable control of deposition speed, current, voltage, and torch path for the benchmark builds.

For the analysis and control of the temperature during the printing process, and as a thermal camera (see Figure 2) mainly gives readings of the surface temperature of the sample, a series of type K thermocouples were spot-welded to the top and bottom of the sample to monitor the temperature inside the specimen (see Figure 2 and Figure 3 B, C and D). A total of 13 thermocouples were used, 8 at the top and 5 at the bottom. Spot-welding, in this context, refers to a method of attaching the thermocouple wires directly to the metal surface using a small electrical current that generates localized heat. It has been used in previous studies such as the work of Peyre et al. on the analytical and numerical modelling of the direct metal deposition laser process [21], and the work of Schnell et al. on experimental and numerical thermal analysis of the laser powder bed fusion process [22]. Other studies, such as the work by Seung Lee [23], have relied on pyrometers to measure the surface temperature of the deposited passes during WAAM. While this approach enables non-contact measurements directly over the melt track, it is highly sensitive to variations in surface emissivity, arc light interference, and the exact positioning of the optical path. These factors often lead to difficulties in capturing the true peak temperatures and can underestimate the rate of heat loss from the part, especially in multi-layer and large-scale builds. Consequently, thermal models calibrated against such surface readings may fail to reproduce the full thermal history, particularly within the bulk of the material where microstructure evolution and residual heat accumulation occur.

Due to the high heat input associated with the WAAM process, managing thermal stresses during deposition is critical, and an effective clamping strategy plays a key role in stabilizing the workpiece. In this setup, four square rods made of nickel aluminium bronze, each with a side length of 2 cm, were employed to secure the base plate (see Figure 2). Two of these rods were positioned on the top surface of the plate, while the other two were placed underneath, creating a symmetrical arrangement that provided mechanical restraint from both above and below. This configuration aimed to limit movement of the base plate during deposition by providing firm contact and resistance against thermal expansion and contraction. The use of nickel aluminium bronze was chosen for its strength and thermal stability, helping to maintain consistent pressure throughout the WAAM process. While the setup was passive and not equipped with active cooling or adaptive force adjustment, it provided a practical and repeatable means of fixturing the base plate for the experimental build. Future work should examine whether active cooling or adjustable fixturing can reduce warping while maintaining comparable thermal histories, especially for larger walls and more industrially relevant geometries.

The final setup with the robotic arm, thermocouples, clamping conditions and sample can be seen in Figure 2.

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Figure 2: Final setup with nozzle and robotic arm, thermal camera, thermocouples, and clamping conditions.

Table 4 summarises the sample and welding parameters used to manufacture all the benchmark specimens:

Table 4: Sample and welding parameters

Property	**316L value**
Density	7.85×10^{-6} kg/mm^3
Arc power	1800 – 2500 W
Plate dimensions	200x100x16 mm
Number of Thermocouples	13 (8 top and 5 bottom)
Number of passes	8 (*2 Layers*), 12 (*3 Layers*)
Interpass temperature	50°C
Voltage	22 V
Current	190 A
Wire-feed speed	6.5 m/min
Arc-length correction	-11%
Printing speed	7 mm/s
Argon flow	0.25 L/s

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Layer height	3.00 mm

In this study, the nozzle was positioned vertically only for the first bead of each layer. For the subsequent three beads, the nozzle was tilted at a controlled angle relative to the print direction (see Figure 3 A). This strategic tilt plays a critical role in improving the quality of the weld and the overall geometry of the deposited structure. From a welder's perspective, tilting the nozzle helps direct the arc force and molten pool more precisely into the side of the previous bead, promoting better wetting action and fusion between adjacent passes. When welding bead-on-plate or multi-pass welds, a tilted torch encourages smoother deposition along the bead interface, reducing the risk of undercutting or lack of sidewall fusion. In WAAM, where overlapping beads are used to build up layers, this approach is especially beneficial to eliminate defects such as voids or cold laps that can form due to insufficient heat or filler penetration at the overlap region. Furthermore, the tilt reduces the shadowing effect where the arc and wire might not fully access the joint between beads if approached perpendicularly. It also helps control bead width and reinforcement, contributing to a more uniform layer height and reducing the likelihood of defect accumulation over successive layers. This practice not only improves metallurgical bonding but also enhances mechanical integrity across the printed structure.

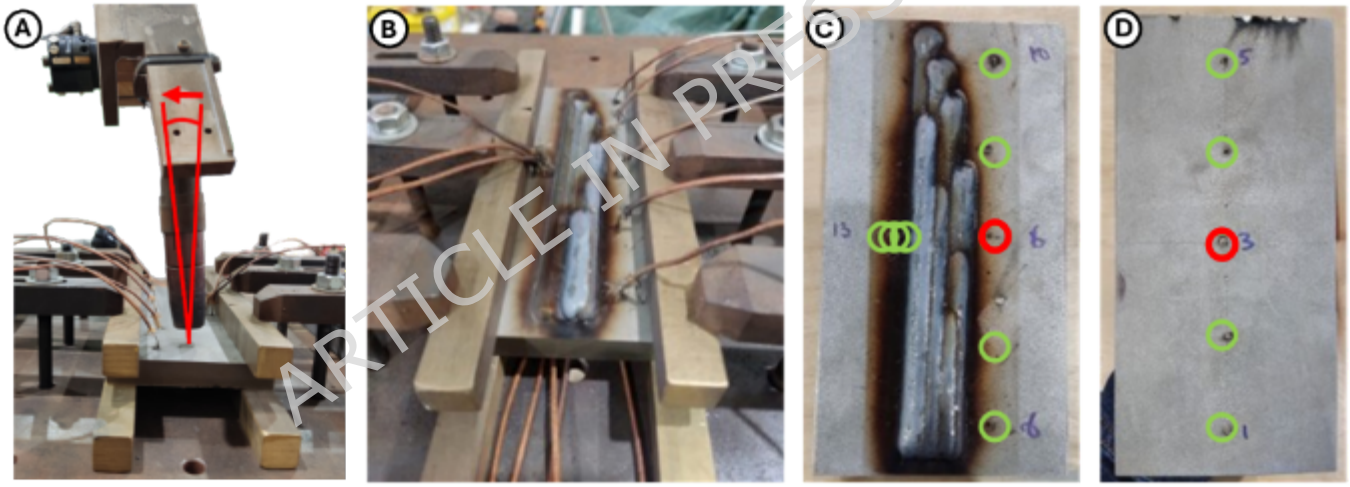


Figure 3: Tilted torch strategy (A) and thermocouple distribution during manufacturing process (B is the general setup and wiring, C is the top view and D is the bottom view of the sample with marked locations of wire connection points).

After completing each pass, the surface of the deposited bead was thoroughly brushed using a stainless-steel wire brush to ensure metal-to-metal contact between layers and hence metallurgical continuity. This step, while simple, is a fundamental practice in high-quality welding and additive manufacturing, especially when working with stainless steel like 316L. The brushing serves multiple purposes. Primarily, it removes oxide layers, spatter, and fine particles that form on the surface during welding due to exposure to heat and air. If these are not removed, they can become trapped between layers or cause inconsistent melting and fusion in subsequent passes. This leads to inclusions, poor bonding, and surface irregularities that compromise structural and aesthetic quality. Additionally, brushing eliminates residual slag or micro-contaminants that may have formed despite shielding gas coverage.

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To maintain consistency and control over the microstructure evolution, the interpass temperature was monitored using embedded thermocouples, ensuring the temperature dropped to approximately 50°C before initiating the next pass. This practice of letting the part cool between passes is crucial in WAAM and welding in general. Allowing the base material to cool reduces the buildup of residual heat, which can otherwise lead to excessive grain growth, distortion, or undesirable phase transformations.

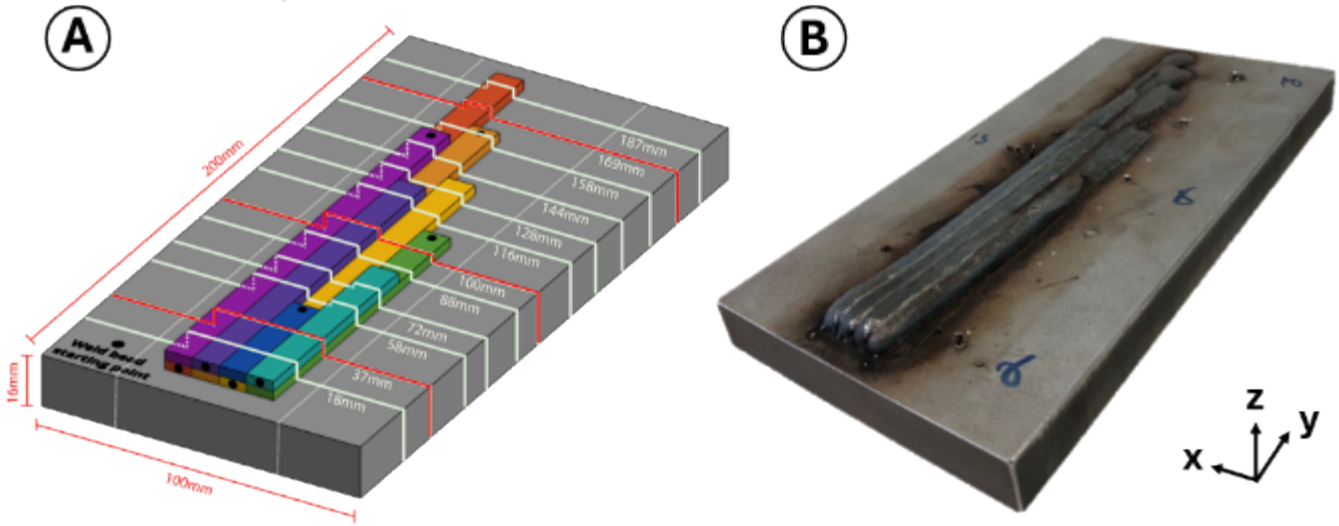
3.3. Manufactured Samples, Cutting and Consumables

To ensure accurate calibration and validation of the simulations, two benchmark specimens were fabricated, each designed to capture different parameters relevant to the developed models. The benchmark specimens are referred to as *2 Layers* and *3 Layers* in this study. The specimens were manufactured on top of a 316L stainless steel base plate with the dimensions of 200x100x16mm, and each layer had four beads/passes. The base plate pieces were cut from the same larger plate, so that the material composition remained consistent throughout the study. Using base plates with nominally identical lengths, widths, depths, and material composition helped to provide a consistent set of results for the comparison between experiments and simulations. The printing pattern followed a zig-zag sequence; (see black dots in Figure 4 A that represents the starting point of the bead).

3.3.1. 2 Layers Sample

The sample with 2 layers (see Figure 4) consists of 8 passes with varying lengths. The distance between the centres of the two passes was 5.5mm and the layer height was 3mm. The decision to have passes with different lengths was motivated by the need to extract cross-sections from various regions of the sample that captured the full range of physical phenomena occurring (e.g. microstructure changes within the passes when subsequent beads are laid) throughout the WAAM process, (see red lines in Figure 4 A representing the cross-sections cut from the sample plate). These included sampling areas from the beginning (ramp-up) and end (ramp-down) of the weld passes, which are typically within the first and last 1.5-2 cm. These transient regions are crucial for future studies aiming to understand the complex thermal and mechanical behaviours during process initiation and termination, where heat input, bead morphology, and material flow differ significantly from the steady-state region.

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Figure 4: 2 Layers sample: A) Design of printing directions and cross sections; B) Isometric view of the manufactured sample.

For the current study of hardness and distortion in the WAAM process, three different cross-sections were extracted from the *2 Layers* sample, and named as follows: Sample 169, Sample 100, and Sample 37. These names correspond to the distance from the front face of the base plate to the corresponding cut (see red lines in Figure 4 A). Sample 169 (Figure 5 A) shows the cross-section of a single pass and served as a calibration for the pass dimensions and shape, as well as the analysis of the area and melt pool. Sample 100 (Figure 5 B) shows the cross-section of 6 passes across 2 layers and their numbering depicting the deposition order in the *2 Layers* sample. The same principle applies to Sample 37 (Figure 5 C), but this time showing 8 passes across 2 different layers. Both Sample 100 and Sample 37 were used for hardness measurements and calibration and validation of the thermomechanical model. These specific positions were chosen because they provide a comprehensive view of the multi-scale interactions inherent in the WAAM process. The cuts include the fusion between the deposited material and the base metal, the overlap between adjacent passes within the same layer, and the bonding between layers. This allows for a detailed examination of all relevant thermal and metallurgical zones, including the weld metal, heat-affected zone (HAZ), and base material.

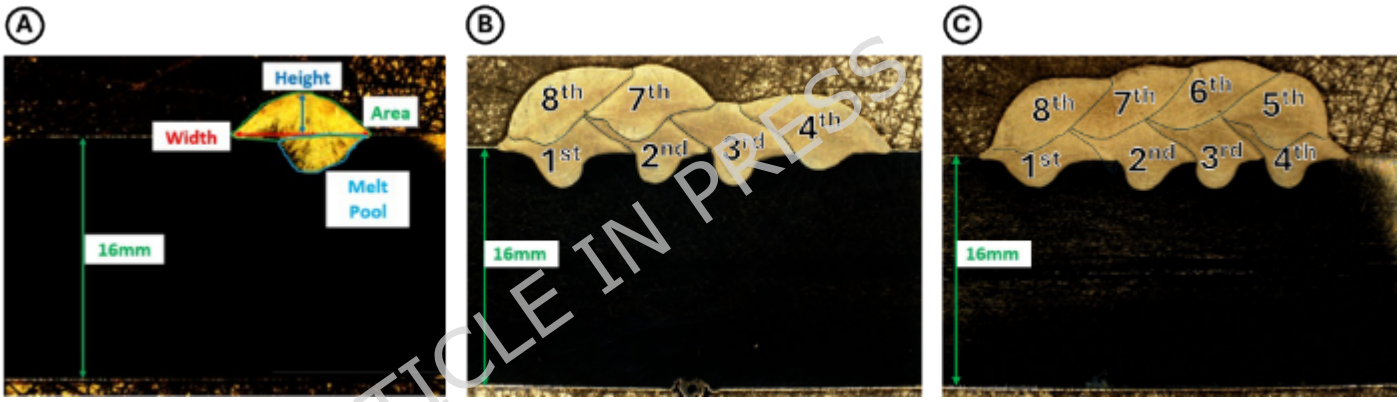


Figure 5: Cross sections of the 2 Layers sample (see Figure 4 A) for the cuts at: A) 169mm; B) 100mm; C) 37mm, from the front of the base plate.

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3.3.2. 3 Layers Sample

The second sample consists of 12 passes, divided in three layers and four passes per layer (Figure 6). For this sample, all the passes were 180mm long, the distance between the centres of two passes was 5.5mm and the layer height was 3mm. The sample was used to calibrate and validate the thermomechanical model (heat losses and temperature evolution during the print, as well as distortion), using the thermocouple recordings on the top and bottom of the specimen, and the measurement of distortion using a GOM System, as described in Section 3.4.

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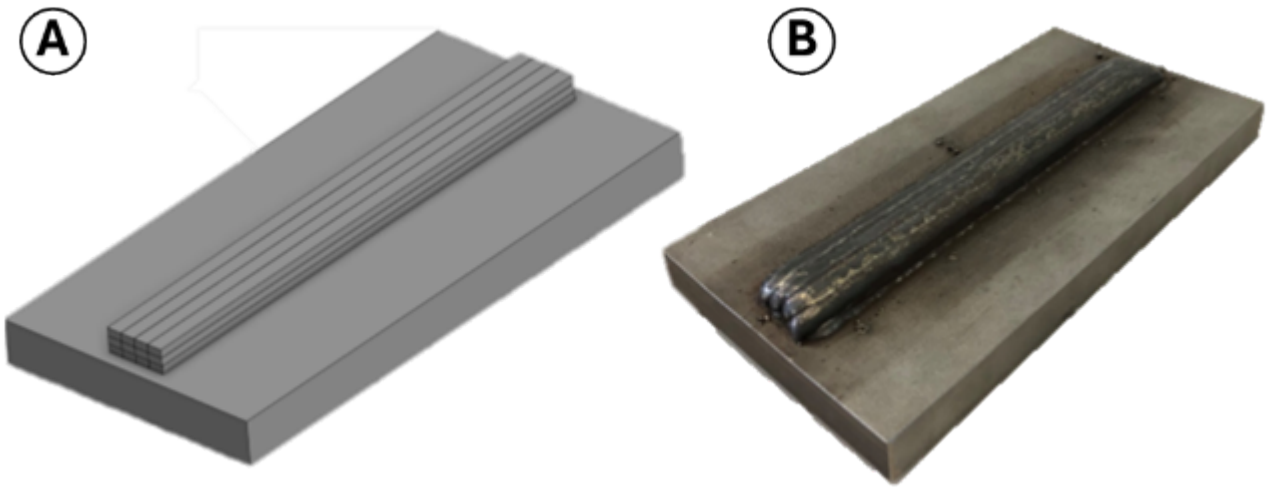


Figure 6: 3 Layers sample in isometric view. A) 3D model; B) manufactured sample.

For the dimension and volume analysis, several software packages (Meshmixer, ImageJ, Cubit, and FreeCAD) were used to obtain an average of the overall volume and size of the passes and the layer height, as well as the overlap between layers and between passes within the same layer.

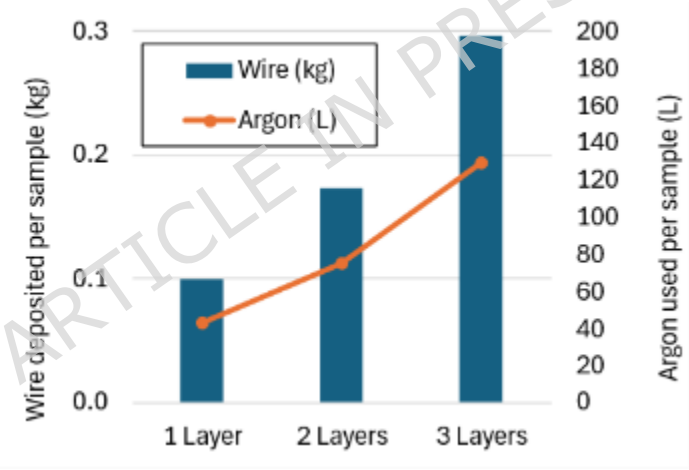


Figure 7: Amount of wire (in kg) and the amount of argon (in L) per sample.

Figure 7 illustrates the amount of wire (in kilograms) and argon (in litres) used for each sample mentioned, and the amount for a 1-layer sample as a nominal reference, which values were obtained during the manufacturing of the *3 Layers* sample. The data highlights the variation in material consumption and gas requirements across the different samples, providing insights into the resource efficiency of the process. This data gives an approximate amount of wire of 138g/m and an Argon flow of 0.25L/s for the WAAM process in this study. This information is critical for assessing the overall material and energy usage in the WAAM process, which can help optimize production costs and improve sustainability.

3.4. GOM System for Distortion Measurements

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Distortion was quantified using a GOM (ZEISS) ATOS 5 structured blue-light system in "Triple Scan" mode at 10 µm lateral sampling. The sensor was calibrated with the manufacturer's artefact, and checks in ZEISS INSPECT [24] verified accuracy. Eight reference markers per face (top and underside) enabled robust multi-view alignment while the specimen was rotated in ~22.5° steps to ensure full coverage. Point clouds were meshed, and the pre-deposition scan was registered to the post-deposition mesh via marker alignment followed by a best-fit constrained to invariant substrate regions. The base-plate geometry was then segmented away to isolate the WAAM build. Surface deviation maps and path-wise displacement profiles were extracted, and distortion was normalised using the warping index (see Eq. 13), defined as the ratio of the vertical deformation (u_z) to the substrate thickness (d):

$$w^* = {}^{u_z}/_{d} \qquad \text{Eq. 13}$$

Uncertainty was mitigated by high view overlap, exposure settings that avoided saturation, and repeated viewpoints for repeatability.

3.5. Hardness Measurements

Vickers hardness was used to characterise the local mechanical response of the WAAM-fabricated 316L samples and to support calibration and validation of the constitutive model. Tests were performed on a Struers DuraScan with automated loading and optical indentation measurement, following ASTM E384 and ISO 6507.

A 3 kgf load (HV3) with a 15 s dwell was applied. The Vickers number (see Eq. 14) was computed from the mean diagonal length

$$HV = \frac{1.8544F}{d^2} \qquad \text{Eq. 14}$$

where F is the applied load in kilograms-force (kgf) and d is the average length of the two diagonals of the indentation in millimetres (mm). Cross-sectioned, polished specimens (final 0.05 µm colloidal silica) were indented across the substrate, heat-affected zone, and deposited passes to capture variations associated with thermal cycling and solidification.

4. Results and Discussion

4.1. Thermal Model Calibration

The area and shape of the different passes was extracted from the cross-sections of the *2 Layers* sample (see Figure 5), and the dimensions were used as input for the simulations. The average pass width was 8.9mm and height 2.6mm, and the average area was 18.4mm^2. The measurement was carried out with ImageJ considering the width of the base plate as scale (16mm), see Figure 5 A. The 8 thermocouples placed on the top surface (see Figure 3 C) of each specimen were strategically distributed to capture both the localized and global thermal response during the WAAM process. Five of these thermocouples were aligned along the printing direction and positioned at a uniform distance from the deposited bead to monitor the thermal behaviour under steady-state conditions during continuous deposition. This alignment allowed for tracking temperature evolution along the path of the heat source and detecting

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potential variations in heat input or deposition consistency. The remaining three thermocouples were spaced laterally, increasingly offset from the deposition track to assess the thermal gradient across the width of the base plate. This arrangement provided valuable data on how heat propagates away from the molten pool and dissipates through the surrounding material, which is crucial for understanding the development of residual stresses and distortion, especially in larger or more complex builds. However, due to their proximity to the heat source, specifically the electric arc, the thermocouples on the top surface were potentially subject to inaccuracies. The arc's intense brightness, along with the radiation from the molten metal, could interfere with the readings by inducing artificial signal fluctuations or elevated temperature readings unrelated to actual conductive heat transfer. These effects are particularly prominent when thermocouples are exposed to direct or reflected arc radiation without adequate shielding.

To address this limitation and obtain more reliable data for calibration, an additional set of five thermocouples was spot-welded on the underside of each specimen (see Figure 3 D). These thermocouples were also aligned in the printing direction and centrally located along the base plate to ensure consistency in the data collection. Being on the opposite side of the plate and shielded from direct arc radiation, these sensors primarily captured the heat transmitted through conduction from the deposition zone, as well as secondary cooling effects due to convection and radiation from the lower surface. As a result, the bottom thermocouples offered a more stable and accurate representation of the actual thermal cycles occurring within the bulk of the material. These measurements were thus selected as the primary dataset for calibrating the thermal boundary conditions in the simulation, including critical parameters such as the heat source efficiency and the surface emissivity governing heat losses to the environment. The accuracy of these bottom measurements was essential in refining the simulation's predictive capabilities, particularly in capturing the thermal field distribution and matching experimental temperature profiles across different deposition scenarios.

Alll thermal boundary conditions and assumptions used in the calibration were: (i) The moving ellipsoidal source delivers a net power $Pf\eta$ with the calibrated efficiency and scaling factor of Table 5. (ii) All exposed external surfaces carry the combined convective–radiative loss of Eq. (2), with a convective coefficient h and emissivity ε calibrated against the underside thermocouples and the radiation term linearised about ambient. (iii) The interface between the base plate and the clamping rods is treated as a local heat-sink condition on the top and bottom plate surfaces beneath the rod footprints: the nickel-aluminium-bronze rods are highly conductive and act as additional conduction/convection paths that draw heat from the plate, so an elevated effective heat-transfer coefficient is prescribed over the contact patches to represent this localised cooling at the base-plate top, rather than treating those surfaces as purely convective to air. (iv) Ambient temperature is fixed and inter-pass dwell is imposed by the temperature-triggered scan advance described in Section 2.4. Two measurement artefacts are explicitly accounted for and are NOT treated as real temperature: spatter impingement on a thermocouple junction produces brief, isolated spikes (occasional fluctuations superimposed on the true history, not a sustained offset), and direct or reflected arc radiation produces a systematic over-reading on the top-surface sensors closest to the arc. For this reason the underside thermocouples, conduction-dominated and shielded from the arc, are used as the primary calibration targets, while the top-surface sensors are used only as qualitative trend checks. The thermocouple readings were then compared to simulation results to assess the accuracy of the heat-transfer parameters implemented in the numerical setup.

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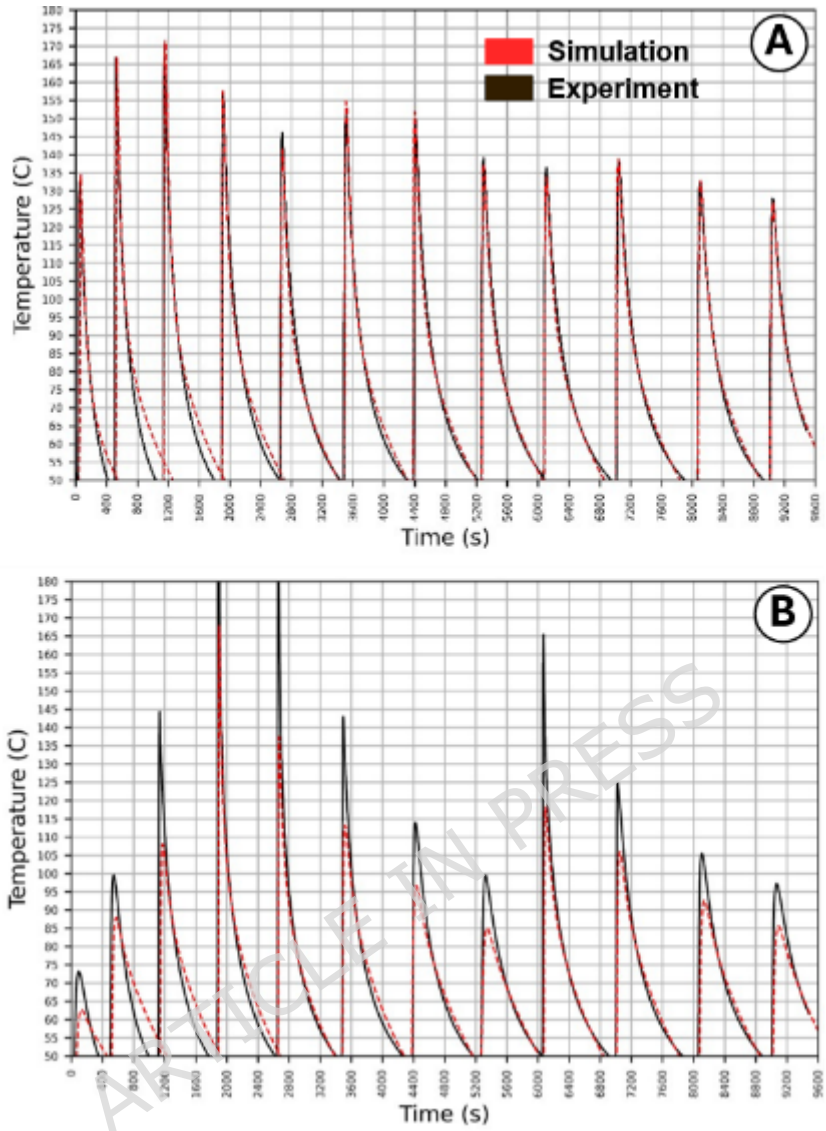


Figure 8: Thermocouple readings from experimental (black line) and simulation (red dashed line) results for thermocouples number 3 (graph A, located at base plate's bottom, see Figure 3 D) and 8 (graph B, located at base plate's top, see Figure 3 C).

The thermocouple readings were compared to simulation results to assess the accuracy of the heat transfer parameters implemented in the numerical setup. Among these, the results from thermocouple number 3 (see red circle in Figure 3 D and Figure 8 A), located at the bottom of the base plate, and thermocouple number 8 (see red circle in Figure 3 C and Figure 8 B), positioned at the top of the deposited structure, are presented here as representative examples. These two thermocouples were selected to capture thermally distinct zones: the former within the substrate acting as a heat sink and the latter near the heat source during deposition. The temperature-time graph for thermocouple number 3 shows a very good agreement between experimental and simulated data in terms of peak temperatures, thermal cycles, and cooling behaviour after calibration. This suggests that the heat losses through the base and the conduction have been accurately represented in the model. This match is critical since the thermal behaviour of the base plate directly affects the initial microstructure, fusion boundary characteristics, and residual stress and distortion formation. The close fit here validates the choice of

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thermal boundary conditions applied in the simulation. On the other hand, thermocouple number 8, positioned on the top of the sample, near the arc, also demonstrates excellent agreement for heat loss and post-deposition cooling behaviour. The measured maximum peaks are systematically higher than the simulation because those sensors are affected by direct arc radiation, spatter and reflected shine, and therefore do not represent pure conductive temperature within the material. For this reason, top-surface thermocouples were used as qualitative checks of thermal trends, whereas the underside thermocouples were used as the primary calibration dataset. A common solution to the influence of the radiant heat from the electric arc to the thermocouples is to apply a protective ceramic coating, which, however, was not used in this study. An overview of the simulated thermal model for the *3 Layers* sample can be seen in Figure 9, where the temperature evolutions for the 1st (Figure 9 A), 4th (Figure 9 B), and 12th (Figure 9 C) pass are depicted.

The semi-axis lengths (a,b,c) of the ellipsoidal source/activation envelope were not assumed a priori but determined from the measured bead geometry and then fine-tuned during the thermal calibration. The procedure was as follows. First, the fusion-zone cross-section of a single calibration pass (Sample 169, Figure 5 A) was measured by optical microscopy with ImageJ, giving an average pass width of 8.9 mm, height 2.6 mm, and fusion area 18.4 mm² (with the 16 mm base-plate thickness as the length scale). These dimensions set the initial transverse and depth semi-axes (b and c), while the longitudinal semi-axis (a) was initialised from the travel speed and the molten-pool length. The values were then adjusted within the bounds set by the measured melt-pool geometry until the simulated effective molten region (the T_liquidus isotherm) reproduced the measured pass width, height and penetration, and simultaneously matched the peak temperatures and cooling rates recorded by the underside thermocouples (TC3). This joint geometric-plus-thermal calibration yielded the final values a = 1.8 mm, b = 1.5 mm, c = 1.5 mm reported below, which are consistent with the measured fusion-zone size. The final calibrated values are shown in Table 5.

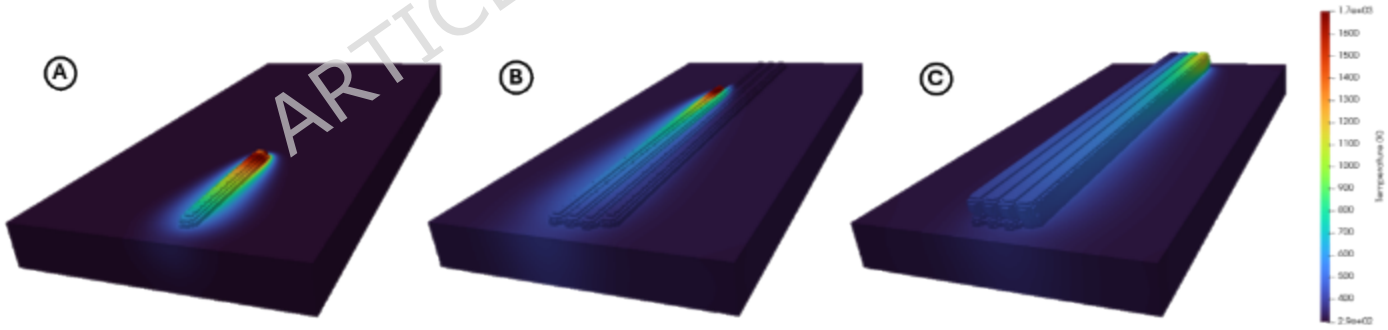


Figure 9: Simulated thermal evolution for the: A) 1st pass; B) 4th pass; C) 12th pass, in the 3 Layers Sample.

The final calibrated values are shown in Table 5.

Table 5: Thermal model calibrated parameters.

Parameter	Calibrated value
Power (P)	1800 – 2500 (in W)
Efficiency (η)	0.70 – 0.77
Scaling factor (f)	1.1 – 1.5
Ellipsoid's Semi Axis Lengths (a,b,c)	1.8, 1.5, 1.5 (in mm)

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Overall, the use of thermocouple readings, particularly the representative examples from thermocouple number 3 and thermocouple number 8, strengthens the reliability of the simulation results and supports the calibration of the thermal model across different regions of the part. As demonstrated by the results from thermocouple number 3, this calibration approach allowed the model to reproduce both the peak temperatures and the cooling curves with high fidelity, providing a realistic representation of the thermal cycles occurring in the material. This, in turn, enhances the predictive accuracy of the model for heat accumulation in multi-layer builds, where the substrate's thermal state plays a critical role in determining hardness and distortion.

4.2. Distortion in WAAM Samples

To further analyse the effect of residual stresses on the sample's final geometry, distortion measurements were taken for each of the specimens after the print. Due to the high heat input characteristic of the WAAM process, substantial thermal gradients arise during deposition, leading to the development of residual stresses that often result in distortion of the part [25], [26], [27], [28].

A critical factor influencing the extent and nature of this distortion is the clamping setup employed during fabrication. Notably, despite the clamping strategy used in this study (Figure 2), distortion was still observed where the thermal stresses induced by successive heating and cooling cycles had a dominant influence. The residual flexibility in the clamping setup, especially in regions away from direct contact with the rods, allowed localized deformation to occur, suggesting that further refinement in fixturing design, potentially involving increased contact area or active cooling, may be necessary to fully control distortion in WAAM-fabricated components [29]. In this study, no pre-heating was applied to the base plate, resulting in a steeper temperature gradient during the deposition process. This significant temperature difference leads to uneven thermal expansion and contraction, ultimately causing warping and distortion as the material cools.

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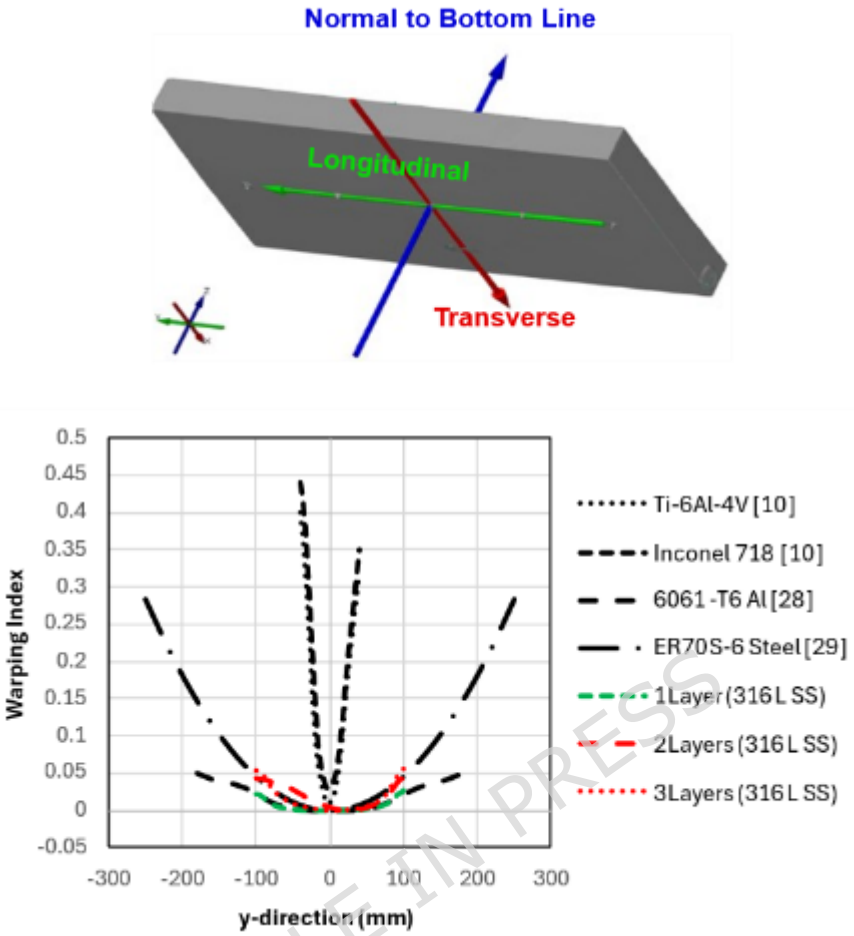


Figure 10: Distortion for longitudinal direction for Ti-6Al-4V [11], Inconel 718 [11], 6061-T6 Al alloy [30], ER70S-6 Steel [31], and 316L SS (current study).

In the present study, distortion was quantified using the warping index, w^*(see Eq.13). This parameter allows comparison between builds made with different plate thicknesses and across different materials, providing a normalized measure of distortion severity. The results showed warping index values of up to 0.06 in the longitudinal direction for the *2 Layers* and *3 Layers* samples (see Figure 10). These values are in reasonable agreement with results from other WAAM studies using steel, such as the work by Huang et al. [31] reporting a warping index between 0.10 and 0.28. Comparisons to other materials show a broad range of warping index values: aluminium alloys typically exhibit a warping index between 0.05 and 0.3 [30], while Ti-6Al-4V shows ≈0.28 [11] and Inconel 718 shows ≈0.37 [11]. These differences also reflect variations in substrate thickness, thermal properties, and deposition strategies used in each study.

The agreement between the present results for 316L and previously reported values suggests that the distortion observed here is within the expected range for this alloy under similar WAAM conditions [11], [30], [31]. The relatively moderate warping index compared to Ti-6Al-4V and Inconel 718 highlights the favourable distortion resistance of 316L, while still indicating the need for careful thermal management to avoid cumulative deformation in multi-layer builds. The use of a normalized metric such as the warping index also facilitates more meaningful comparisons across different materials and

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experimental setups, providing a clearer context for interpreting the mechanical restraint and thermal control strategies required in future WAAM process optimization [32].

The *3 Layers* sample with uniform pass length and pattern shows an almost perfectly symmetric behaviour of the warping index along the y-direction (see Figure 11 D). Additionally, a 1-layer sample was simulated to showcase the capabilities of this model (see Figure 11 C-D), and the distortion results are within the expectations, i.e., the 1-layer sample shows a smaller distortion in both directions as only 4 passes were deposited in a single layer. On the other hand, the *2 Layers* sample clearly displays the asymmetry in the y-direction, with a higher warping index observed on the side of the sample where the longer passes create a higher thermal cycle (see Figure 11 B). This behaviour is also predicted by the simulations for both transverse and longitudinal directions (see Figure 11 A-B).

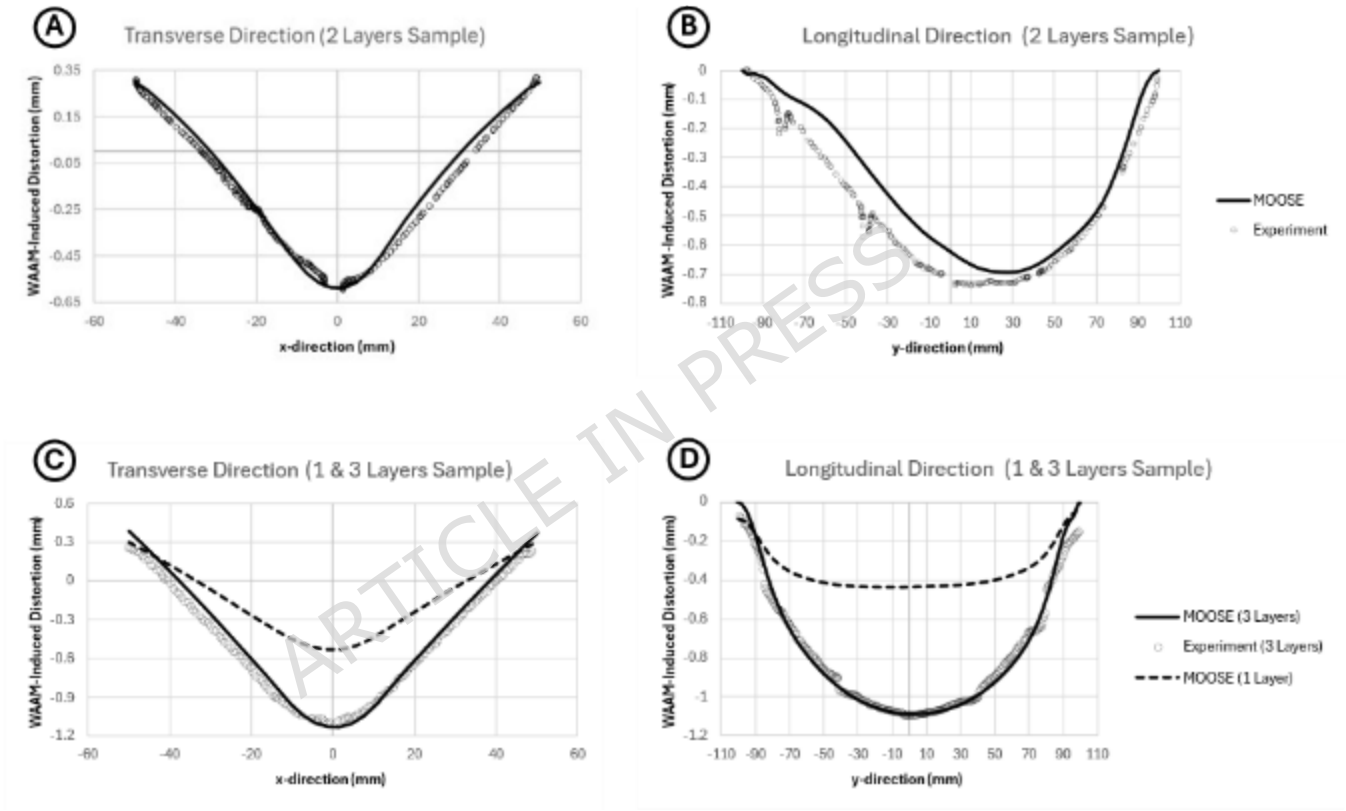


Figure 11: WAAM-Induced distortion for longitudinal and transverse directions through the bottom of the base plate for experimental results and compared with thermomechanical simulations (MOOSE).

Table 6 shows the errors in the distortion predictions (in mm) depicted in Figure 11. These values fall inside the acceptable ranges previously reported by other studies such as the work of Wacker et al. [33], where they used Artificial Neural Networks for geometry and distortion prediction, or Prajadhiana et al. [34] who used Enhanced Separation and Deposition Algorithms. Both works reported errors between 0.035 and 2.03 mm. These errors are affected by many factors, like substrate plate thickness and shape, and the constitution law in the finite element model.

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Table 6: Absolute distortion error (in mm) for each sample and direction.

Sample	Direction	Absolute Distortion Error (mm)
2 Layers	Longitudinal	0.103
	Transverse	0.088
3 Layers	Longitudinal	0.037
	Transverse	0.062

4.3. Hardness in WAAM Samples

Microhardness mapping was carried out on Sample 37 (see Figure 5 C) and Sample 100 (see Figure 5 B), and the corresponding hardness fields were also predicted numerically from the accumulated equivalent plastic strain fields obtained from the calibrated thermomechanical models implemented in the MOOSE Framework (Figure 13 A-D)). For WAAM 316L, the calibrated parameter set (see Eq. 12) used in this work was A=140, B=320 and C=0.45. A single set of parameters A, B and C is used for all regions (FZ, HAZ and base metal). This choice reflects the fact that all zones are composed of fully austenitic 316L and are modelled with a single temperature-dependent constitutive law; differences in local yield strength and work-hardening behaviour are captured through the heterogeneous thermal histories and accumulated plastic strains predicted by the thermomechanical analysis, rather than through region-specific hardness laws. Physically, the assumption is that, after cooling to room temperature, the local hardness is primarily controlled by the stored plastic deformation (dislocation density) in the austenitic matrix, so that a unique hardness–plastic strain relation is appropriate for the alloy as a whole [35].

For the selection of the final values, a parametric study on the Full Model was performed in which the three coefficients were varied over physically plausible intervals (see Figure 12) and compared to the V1 experimental line (see Figure 13 E). In particular, A was explored between 100 and 170 HV, B between 280 and 380, and C between 0.2 and 0.6. These trials showed that low values of A underestimate hardness in the parent metal and therefore reduces the overall value of the hardness, while higher values overpredict the minimum and peak hardness. Lower values of B underpredicts the maximum peak of hardness, and the opposite happens for high values of B. Exponents C<0.45 lead to high values of hardness and creates a major difference between minimum and maximum values of hardness, whereas C>0.45 produces excessively flattened hardening bands and lower peak values that are not observed experimentally. The lowest error is given by the selected values of these parameters, with an average error of 8HV (see Figure 12). The adopted set provides the best compromise between absolute levels and spatial gradients of hardness across all sections.

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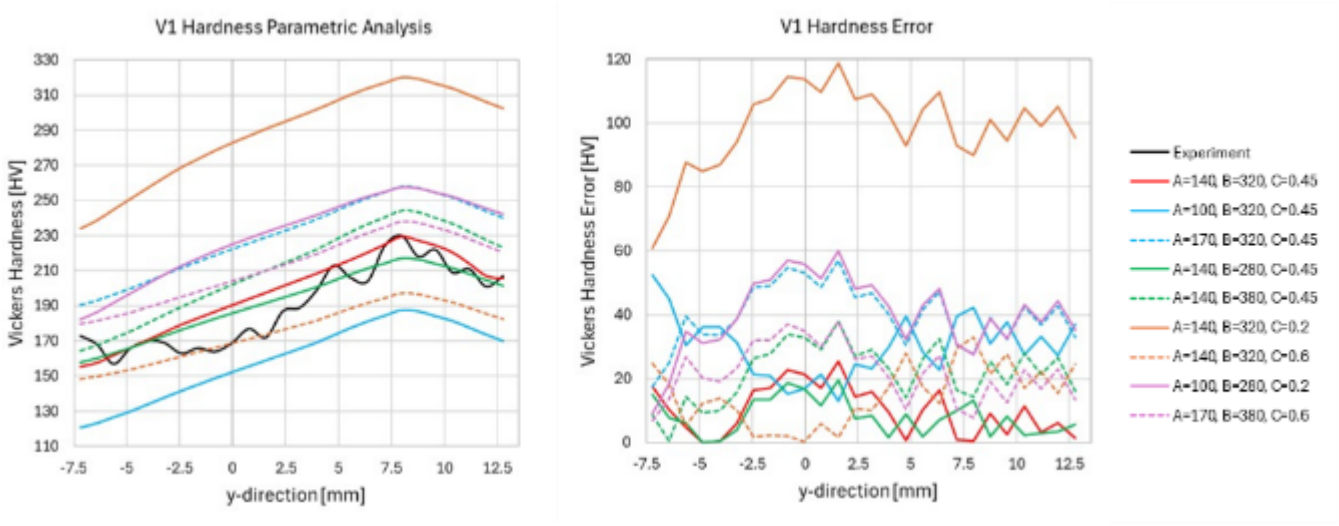


Figure 12: Hardness parametric analysis through V1 line and error between experimental and simulation results.

For each model (Simple and Full) and for the experimental hardness map, a total of 3 lines (see V1, V2, and H1 in Figure 13 A, C, E) were studied to show the accuracy of the model and the evolution of the hardness during the WAAM process. V1 goes through the centre of the cross-section along the normal direction. V2 follows the same direction but it is situated close to the edge of the printed wall, to show differences in the hardness at those areas. H1 represents the fusion line, where the first layer is deposited on top of the base plate.

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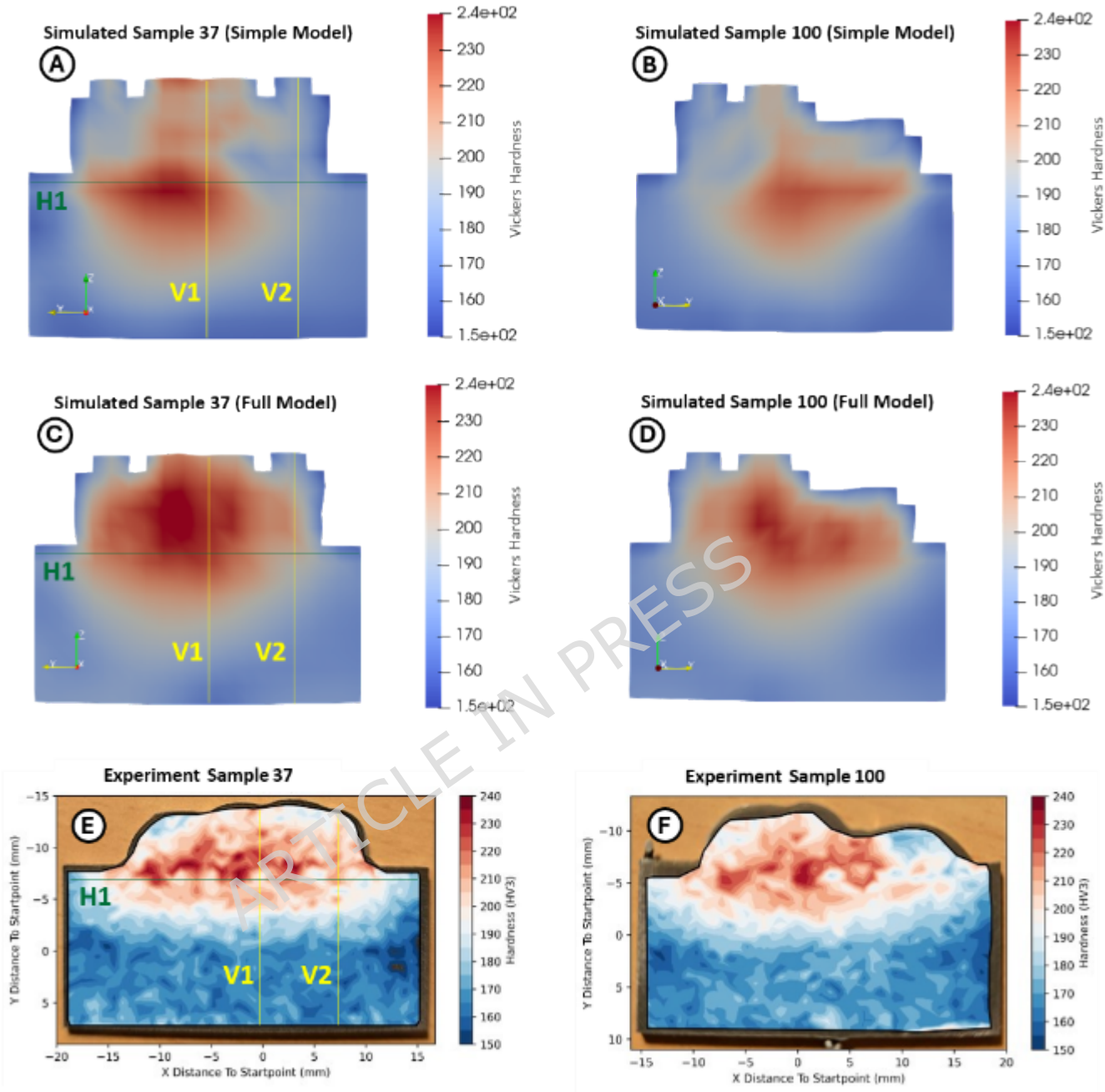


Figure 13: Hardness values for: A-B) Simple Model; C-D) Full Model; E-F) Experiments, for both cross-sections at 37mm and 100mm.

The hardness map for Sample 37 (see Figure 13 A, C, E) shows a consistent increase in hardness in the weld metal and HAZ, with values in the range of 200 to 240 HV at the fusion boundary and interface between the deposited passes and the base plate, in the overlapping regions between adjacent passes within the wall, and in the HAZ immediately below the wall in the base plate. Away from the HAZ, the base plate hardness stabilises between 150 and 180 HV, consistent with solution-annealed 316L stainless steel. The through thickness profile therefore exhibits a clear gradient: the softest material is the unaffected base metal, hardness increases in the HAZ where thermal cycles and constraint promote

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plastic straining, and the highest hardness is found within the deposited passes and their overlap regions, where repeated heating and cooling and the associated thermal mismatch induce the largest accumulated plastic strains. The hardness values for V1, V2, and H1 can be seen in Figure 14. The Simple Model shows a variability of the hardness of ±30HV, while the Full Model decreases this error to ±8HV on average, which is in the range of the machine error itself, associated with local microstructural heterogeneity and indentation scatter.

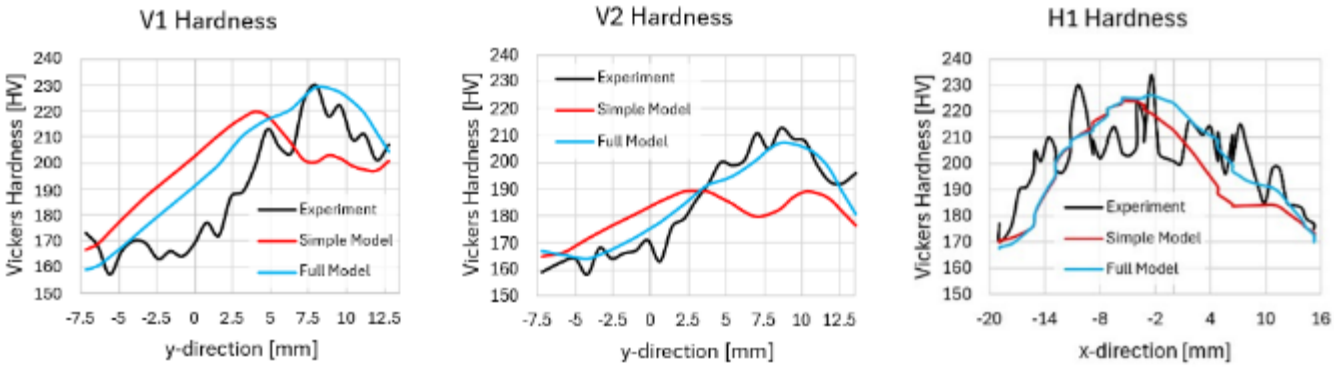


Figure 14: Hardness profile through V1, V2, and H1.

For the Simple Model (see Eq. 10 and Figure 13 A-B), the simplified formulation systematically misrepresented the hardness contrast between overlapping passes and the surrounding material and underpredicted hardness in regions where shear dominated plasticity is expected. This behaviour indicates that shear strains make a non-negligible contribution to the local work hardening in WAAM 316L. The comparison confirms that a full J_2 based equivalent plastic strain (see Eq. 11 and Figure 13 C-D), which combines both normal and shear components in an invariant manner, is required to obtain realistic hardness fields from thermomechanical simulations in this context. Some discrepancies remain at smaller scales. The simulated hardness gradients tend to be more diffuse than in the experimental maps, especially near the fusion boundaries where the measurements show sharper transitions and local maxima reaching 230 to 240 HV3. This behaviour is expected, given that the hardness model depends only on the accumulated equivalent plastic strain and does not explicitly include local microstructural features such as dendrite arm spacing, sub grain structures or local residual delta ferrite content, all of which can modulate hardness at the scale of individual indentations.

For Sample 100 (see Figure 13 B, D, F), the asymmetric deposition pattern is clearly reflected in the hardness distribution. On the left, where both first and second layer passes are present, hardness again reaches 200 to 240 HV within the weld metal and the underlying HAZ, similar to Sample 37. On the right-hand side, where only the first layer is present and the second layer has not yet been deposited, the hardness in the wall and HAZ progressively decreases to approximately 180 to 190 HV. Regions that experience stronger, more constrained thermal-mechanical cycling during deposition acquire larger plastic strain and therefore higher hardness at the end of cooling, whereas regions that are free to relax or that see milder reheating accumulate less strain and remain softer. Along bead flanks and free surfaces, the lateral constraint is weaker, so thermal expansion and contraction are partially accommodated elastically rather than plastically. In addition, edge regions sit outside the strongest overlap of subsequent tracks and therefore experience fewer high-peak, high-gradient thermal cycles. When they are reheated, the stress state is closer to plane stress and peak temperatures often promote recovery without driving

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significant additional plasticity. The combination of easier strain relief and milder reheating limits the buildup of equivalent plastic strain at the edges compared with interior overlap bands, producing a persistent dip in hardness near bead toes and wall flanks. The base plate beneath the HAZ exhibits hardness values in the 150 to 180 HV range for both sections, indicating that the bulk of the substrate remains close to its original condition away from the fusion boundary. Taken together, these two sections illustrate the strong sensitivity of local hardness to the number of thermal cycles and the level of mechanical constraint. Regions that have experienced two deposition passes (first and second layer) and therefore more intense and repeated thermal mismatch show systematically higher hardness than regions that have only undergone a single deposition pass. The fact that the absolute levels remain moderate (about 200 to 240 HV) is consistent with the austenitic microstructure of 316L and the absence of martensitic transformations, which indicates that the hardness increase is dominated by plastic strain accumulation and work hardening rather than phase changes.

To further show the capabilities of this model, we can now analyse the evolution of the hardness within V1, V2 and H1 versus time.

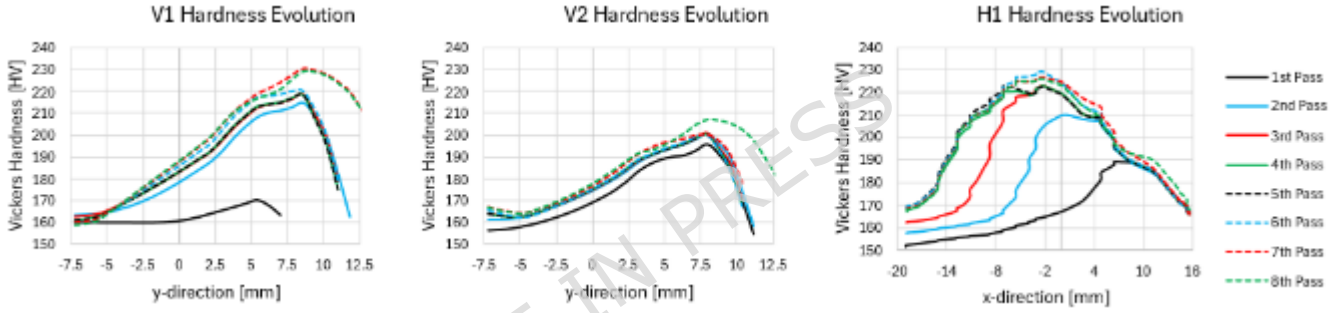


Figure 15: Hardness evolution per pass for V1, V2, and H1.

Figure 15 shows the evolutions of the hardness along V1, V2, and H1 as successive passes are deposited during the WAAM process for Sample 37 corresponding to the *2 Layers* sample. When a new pass is laid, the material beneath and immediately adjacent to the moving pool expands against already solid surroundings, then contracts during cooldown. This sequence increases the local equivalent plastic strain both in the fresh bead and in the material directly beneath it. The largest increment occurs *in column*, that is, in the bead of the lower layer that lies exactly under the centreline of the newly deposited pass. There the peak temperature and through-thickness gradient are highest and the expansion is most strongly constrained, so additional plastic straining is greatest. With each added layer, these in-column bands increase in hardness. After several layers, the mid-height of the wall, which has undergone the most repeated, constrained reheats, exhibits the highest hardness, while zones that have been reheated fewer times or with less constraint show smaller increases.

Following this model validation, we are now able to analyse and study the entire 3D hardness map for any component. Figure 16 shows hardness for longitudinal sections extracted through the interior of the *2 Layers* sample for the first pass (Figure 16 A-C), for the last pass (Figure 16 D-F), and the final full 3D map (Figure 16 G).

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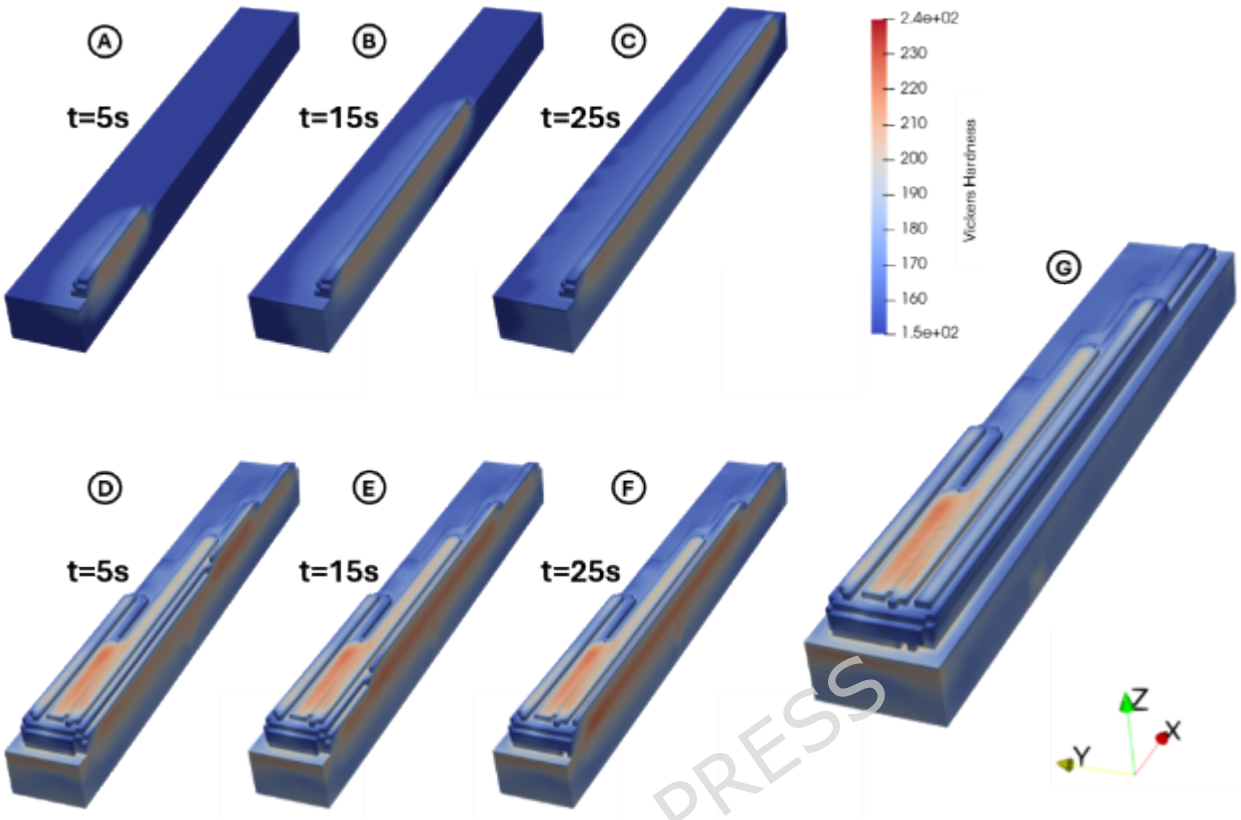


Figure 16: 3D hardness maps for the 2 Layers Sample. A-C (first pass); D-F (last pass); G (full 3D simulation).

4.4. Discussion

Because hardness is linked to the accumulated equivalent plastic strain through the power law described in Section 2.5, the evolution of HV closely follows the development of plastic strain as the wall grows. After the first pass is deposited (Figure 16A-C), the longitudinal section reveals three distinct regions. The bulk of the base plate retains a low and nearly uniform hardness, indicating that most of the substrate remains elastically loaded. A narrow band of increased hardness appears in the heat affected zone beneath the fusion line. Above this, the weld metal in the first bead exhibits still higher hardness, with the largest values concentrated near the weld to plate interface, while the bead crown is slightly softer. This pattern reflects the constraint provided by the thick substrate. During cooling, the thermal contraction of the bead is resisted primarily at the fusion boundary, which accumulates the highest plastic strains and therefore the highest hardness, whereas both the remote base plate and the free surface at the top of the bead experience lower constraint and remain comparatively softer.

When the second layer (last pass) is deposited (Figure 16 D-F), the hardness field evolves in a way that makes the action of the thermal cycle particularly clear. The three-dimensional maps show that the region between the first and second layers develops a new hardness maximum, indicating additional plastic straining in the interlayer zone as the underlying material is reheated and again forced to contract against surrounding constraint. The hardness in this interlayer region exceeds that of the first pass alone and creates a continuous band of elevated HV that follows the layer interface along the wall. The overall result (Figure 16 G) is a through-thickness distribution where the highest hardness is located at layer interfaces and along the fusion line, whereas the interior of the base plate and the free surfaces remain

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closer to the annealed hardness level. This behaviour is consistent with the competing effects of work hardening and local thermal softening in multi-pass welding and illustrates how repeated thermal cycles redistribute hardness within the wall. From a practical viewpoint, the observed hardness gradients are important because they indicate non-uniform local strengthening within the wall. Regions with elevated hardness at overlap bands and fusion boundaries are also the regions that have experienced the strongest thermal-mechanical cycling and the highest strain accumulation. In service, such gradients may influence local yielding, strain redistribution, machining response, and the consistency of mechanical performance across the component. The ability to predict these gradients numerically is therefore relevant for process qualification and for tailoring WAAM strategies toward target local properties.

These three-dimensional simulations provide information that is difficult to obtain experimentally through traditional hardness testers, since accessing the interior of the wall would require extensive and destructive sectioning. In practice, hardness is generally measured only on the cut surfaces after wire cutting, where the local stress state at the indentation site may differ from that in the intact component represented in the modelling. By resolving the full hardness field at any stage of the build, the model allows the effect of deposition order, path strategy, interpass temperature and clamping conditions on the local mechanical response to be analysed in a systematic way. This capability represents a step beyond most existing WAAM studies, which typically restrict validation to temperature histories or global distortion. Here, hardness is treated as a derived quantity tied to the underlying plastic strain field, so that once the thermal and mechanical models are calibrated, the same framework can be used to predict property gradients for new process conditions. Hardness gradients imply local strength gradients, and therefore non-uniform deformation response. In practical terms, this opens the door to using open-source thermomechanical simulations not only to match existing experiments, but to tailor WAAM processing parameters to target specific hardness distributions and, by extension, local strength and stiffness profiles in future component designs.

The present methodology is intentionally focused on wall-type 316L benchmarks and on a small-strain thermo-elasto-plastic framework with a strain-based hardness surrogate. Although this is sufficient for the present calibration and validation tasks, industrial components may require more complex path planning, larger domains, more detailed clamping models, and, for other alloys, phase-transformation-aware constitutive descriptions. Start and stop transients are also not yet represented with bead-specific heat-source parameters. Nevertheless, the present benchmark strategy provides a scalable foundation because the calibration logic, measurement layout, and open-source workflow can be transferred to more complex builds [36].

5. Conclusions

(1) We introduce two purpose-built WAAM 316L benchmarks (2- and 3-layer walls) that isolate transient, steady-state and overlap effects in a single print and are tailored to joint validation of thermal fields, distortion and hardness.

(2) Thermal calibration using a top and bottom thermocouple layout yields simulations that match measured peak temperatures and cooling histories; bottom sensors provide conduction-dominated, arc-insensitive references that anchor the boundary conditions.

(3) Distortion quantified via a normalised warping index shows clear dependence on layer count and pass pattern; MOOSE thermo-mechanical simulations reproduce longitudinal and transverse distortion with absolute errors ≤0.10 mm and magnitudes consistent with literature for 316L.

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(4) A single power-law mapping of accumulated equivalent plastic strain to Vickers hardness predicts full-field 3D hardness: ≈150-180 HV in the base plate, increasing to ≈200-240 HV at the fusion boundary, layer interfaces and overlap bands, with persistent softening near free edges.
(5) The "Simple" hardness surrogate (normal strains only) misses overlap-band hardening and edge softening; the "Full" J_2-invariant (including shear and volumetric correction) accurately recovers experimental contrasts and spatial gradients.
(6) Together, the benchmarks, datasets, and open-source MOOSE workflow (with progressive element activation) provide a reusable testbed to validate heat-source models, thermal boundary conditions, deposition strategies, and strain-to-hardness laws, and to extend validation to residual-stress and microstructure predictions.

Data availability

The raw/processed data required to reproduce these findings cannot be shared at this time as the data also forms part of an ongoing study.

CRediT authorship contribution statement

Fernando Valiente-Dies: Writing – original draft, Visualisation, Methodology, Conceptualisation, Investigation, Data curation

Nicolò Grilli: Conceptualisation, Investigation, Writing – review & editing

Vladislav Yakubov: Data processing, Writing – review & editing

Joseph Polden: Resources, Writing – review & editing

Tao Wei: Writing – review & editing, Data processing

Kim J.R. Rasmussen: Writing – review & editing, Funding acquisition, Supervision

Anna M. Paradowska: Writing – review & editing, Conceptualisation, Project administration, Funding acquisition, Supervision

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6. References

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[1] G. Knapp, J. Coleman, M. Rolchigo, M. Stoyanov and A. Plotowski, “Calibrating uncertain parameters in melt pool simulations of additive manufacturing,” *Computational Materials Science,* vol. 218, p. 111904, 2023.

[2] N. Patil, R. Ganeriwala, J. Solberg, N. Hodge and R. Ferencz, “Benchmark multi-layer simulations for residual stresses and deformation in small additively manufactured metal parts,” *Additive Manufacturing,* vol. 45, p. 102015, 2021.

[3] NIST, “AM Bench 2025,” 2025. [Online]. Available: https://www.nist.gov/ambench.

[4] H. Jin, J. Long, Y. Kong, J. Chu, M. Wang, X. Liu, W. Zhao, F. Liu, Y. Han and Y. Wei, “Mitigating microstructural and mechanical heterogeneity in WAAM-fabricated Ni-Cr-Mo-based superalloy via post-heat treatment,” *Journal of Materials Processing Technology,* vol. 346, p. 119115, 2025.

[5] A. Horgar, H. Fostervoll, B. Nyhus, X. Ren, M. Eriksson and O. Akselsen, “Additive manufacturing using WAAM with AA5183 wire,” *Journal of Materials Processing Technology,* vol. 259, pp. 68-74, 2018.

[6] A. Caballero, J. Ding, S. Ganguly and S. Williams, “Wire + Arc Additive Manufacture of 17-4 PH stainless steel: Effect of different processing conditions on microstructure, hardness, and tensile strength,” *Journal of Materials Processing Technology,* vol. 268, pp. 54-62, 2019.

[7] J. Goldak, A. Chakravarti and M. Bibby, “A new finite element model for welding heat sources,” *Metall Trans,* vol. 15, p. 299–305, 1984.

[8] X. Liang, L. Cheng, Q. Chen, Q. Yang and A. To, “A Modified Method for Estimating Inherent Strains from Detailed Process Simulation for Fast Residual Distortion Prediction of Single-Walled Structures Fabricated by Directed Energy Deposition,” *Additive Manufacturing,* vol. 23, pp. 471-486, 2018.

[9] K. Prajadhiana, Y. Manurung, M. Fateri, H. L. Choo, W. E. Wan Abdul Rahaman, M. S. Adenan, H. Ambarita, B. Olanrewaju Yusuf, D. Ishak, T. Taufek, M. Ishak, F. Yusof and Triyono, “Distortion analysis of WAAM component using thermo-mechanical, inherent strain and experimental methods,” *Progress in Additive Manufacturing,* vol. 10, pp. 8139-8155, 2025.

[10] R. Sampaio, J. Pragana, I. Bragança, C. Silva, C. Nielsen and P. Martins, “Modelling of wire-arc additive manufacturing – A review,” *Advances in Industrial and Manufacturing Engineering,* vol. 6, p. 100121, 2023.

[11] Q. Wu, T. Mukherjee, C. Liu, J. Lu and T. DebRoy, “Residual stresses and distortion in the patterned printing of titanium and nickel alloys,” *Additive Manufacturing,* vol. 29, p. 100808, 2019.

ARTICLE IN PRESS

[12] Z. Li, M. Bellet, C.-A. Gandin, M. V. Upadhyay and Y. Zhang, “Metallurgically-driven thermomechanical analysis of multiple side-to-side laser melting on a 316L substrate,” *Additive Manufacturing,* vol. 112, p. 104991, 2025.

[13] D. Tabor, The Hardness of Metals, Clarendon Press, 1951.

[14] F. O. Sonmez and A. Demir, “Analytical relations between hardness and strain for cold formed parts,” *Journal of Materials Processing Technology,* 2007.

[15] G. Sundararajan and Y. Tirupataiah, “The hardness-flow stress correlation in metallic materials,” *Bull. Mater. Sci,* vol. 17, pp. 747-770, 1994.

[16] C. J. Permann, "MOOSE: Enabling Massively Parallel Multiphysics Simulation," *Nov,* no. v, 11 2019.

[17] D. Weisz-Patrault, P. Margerit and A. Constantinescu, “Residual stresses in thin walled-structures manufactured by directed energy deposition: In-situ measurements, fast thermo-mechanical simulation and buckling,” *Additive Manufacturing,* vol. 56, 2022.

[18] F. Dunne and N. Petrinic, “Introduction to Computational Plasticity,” *Oxford University Press on Demand,* 2005.

[19] C. Xia, Z. Pan, J. Polden, H. Li, Y. Xu, S. Chen and Y. Zhang, “A review on wire arc additive manufacturing: Monitoring, control and a framework of automated system,” *Journal of Manufacturing Systems,* vol. 57, pp. 31-45, 2020.

[20] ABB, “ABB IRB 2600,” 2025. [Online]. Available: https://new.abb.com/products/robotics/robots/articulated-robots/irb-2600.

[21] P. Peyre, P. Aubry, R. Fabbro, R. Neveu and A. Longuet, “Analytical and numerical modelling of the direct metal deposition laser process,” *Journal of Physics D: Applied Physics,* vol. 41, p. 025403, 2008.

[22] N. Schnell, M. Schoeler, G. Witt and S. Kleszczynski, “Experimental and numerical thermal analysis of the laser powder bed fusion process using in situ temperature measurements of geometric primitives,” *Materials & Design,* vol. 209, p. 109946, 2021.

[23] S. Lee, “CMT-Based Wire Arc Additive Manufacturing Using 316L Stainless Steel: Effect of Heat Accumulation on the Multi-Layer Deposits,” *Metals,* vol. 10, p. 278, 2020.

[24] Zeiss, “OPTICAL 3D SCANNING ATOS 5,” Zeiss, [Online]. Available: https://www.zeiss.com/metrology/en/systems/optical-3d/3d-scanning/atos/atos-5.html.

[25] L. Nguyen, J. Buhl, R. Israr and M. Bambach, “Analysis and compensation of shrinkage and distortion in wire-arc additive manufacturing of thin-walled curved hollow sections,” *Additive Manufacturing,* vol. 47, 2021.

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[26] Y. Manurung, K. Prajadhiana and M. Adenan, “Analysis of material property models on WAAM distortion using nonlinear numerical computation and experimental verification with P-GMAW,” *Archiv.Civ.Mech.Eng,* vol. 21, p. 32, 2021.

[27] M. Gouge and P. Michaleris, “Distortion in metal Additive Manufacturing: Modelling and mitigation,” 2017. [Online]. Available: https://www.metal-am.com/articles/distortion-in-metal-3d-printing-modelling-and-mitigation/.

[28] T. Rodrigues, V. Duarte, R. Miranda, T. Santos and J. P. Oliveira, “Current Status and Perspectives on Wire and Arc Additive Manufacturing (WAAM),” *Materials,* vol. 12, p. 1121, 2019.

[29] K. Prajadhiana, T. Taufek, W. E. Wan Abdul Rahaman, Y. Manurung, M. S. Adenan, M. F. Mat, M. Fuzi, M. Nasir, M. Ackiel, F. Jamaludin and T. Triyono, “Distortion Analysis Method for Wire Arc Additive Manufacturing Component Using Thermomechanical Computation with Enhanced Separation and Deposition Algorithm,” *3D Printing and Additive Manufacturing,* vol. 11, 2023.

[30] Y. Han, “A Finite Element Study of Wire Arc Additive Manufacturing of Aluminum Alloy,” *Applied Sciences,* vol. 14, p. 810, 2024.

[31] H. Huang, N. Ma, J. Chen, Z. Feng and H. Murakawa, “Toward large-scale simulation of residual stress and distortion in wire and arc additive manufacturing,” *Additive Manufacturing,* vol. 34, p. 101248, 2020.

[32] S. Ahmad, Y. Manurung, K. Prajadhiana, B. Olanrewaju Yusuf, M. F. Mat, N. Muhammad, M. Leitner and S. Saidin, “Numerical modelling and experimental analysis on angular strain induced by bead-on-plate SS316L GMAW using inherent strain and thermomechanical methods,” *The International Journal of Advanced Manufacturing Technology,* vol. 120, 2022.

[33] C. Wacker, M. Köhler, M. David, F. Aschersleben, F. Gabriel, J. Hensel, K. Dilger and K. Dröder, “Geometry and Distortion Prediction of Multiple Layers for Wire Arc Additive Manufacturing with Artificial Neural Networks,” *Applied Sciences,* vol. 11, p. 4694, 2021.

[34] P. KP, T. T, W. A. R. WE, M. YHP, A. MS, M. MF, M. F. MFAB, M. N. MAB, M. MA, J. MF and Triyono, “Distortion Analysis Method for Wire Arc Additive Manufacturing Component Using Thermomechanical Computation with Enhanced Separation and Deposition Algorithm,” *3D Print Addit Manuf.,* vol. 11, 2024.

[35] L. Yu, K. Nishimoto and H. Hirata, “Hardness Prediction of the Heat-Affected Zone in Multilayer Welded SUS316 Stainless Steel Based on Dislocation Density Change Behavior,” *Metall Mater Trans A,* vol. 55, p. 1788–1803, 2024.

[36] S. Ghazali, M. Ibrahim and Y. Manurung, “A focused review on numerical computation in wire arc additive manufacturing for high strength low alloy steels: past insights and potential opportunities,” *Int J Adv Manuf Technol,* vol. 141, p. 2679–2708, 2025.